

Interlocking Directorates and Innovation

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Abstract

This paper quantifies how shared board members at competing companies affect innovation. To account for persistent differences in board member ability, I introduce a new empirical strategy that relies on using estimated board member fixed effects from the AKM model as control variables in a second stage. I show that board interlocks are associated with higher innovative output, which is matched by a parallel increase in cross citations between the two interlocked partners. However, this increase in patent counts is concentrated in continuation patents, which are associated with strategic patenting motives. Furthermore, the cross citations go increasingly towards partner patents that are older, more prominent, and not technologically proximate to the citing patent. Therefore, while board interlocks do increase patents, this paper points to strategic patenting behavior being the dominant mechanism as opposed to technological learning.

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1 Introduction

Do shared board members at competing firms influence innovation incentives? If so, through which channels? Because innovation requires large investments in R&D and produces knowledge that can spill over across firms, interlocking boards could either dampen innovation incentives (by reducing rivalry) or enhance them (by sharing information and reducing duplicative R&D). Early seminal work formed theories of why interlocks may form, which gives us a window into why they may also affect innovation. Resource dependence theory views boards as boundary-spanning mechanisms through which firms manage environmental uncertainty and dependence by obtaining advice, legitimacy, and access to external support (Pfeffer, 1972, Pfeffer and Salancik, 1978). A related cooptation literature sees interlocks as links through which firms manage external constraints and stabilize relations with counterparties (Burt, 1983). Reciprocity and social cohesion models emphasize that interlocks may reflect mutually beneficial ties among firms and directors, helping firms exchange information, reputation, and influence (Allen, 1974, Mizruchi, 1996). These mechanisms imply that interlocks may reduce uncertainty and facilitate information exchange (Stein, 2008), but may also support coordination among rivals (Herrera-Caicedo, Jeffers and Prager, 2026). Battaglion and Cerasi (2020) formalize a strategic version of this logic in an oligopoly model, showing that firms may invite executives from rival companies onto their boards to disclose private cost information. However, despite this rich background, there is no consensus on whether interlocks impact innovation in a positive or negative direction.

This paper addresses the gap in the literature. The way it does it is by introducing a new way to account for non-random director assignment across firms. As an intermediate step, I estimate an AKM model (Abowd, Kramarz and Margolis, 1999) for my sample of board members and firms, and then use the resulting board member fixed effects as controls for latent director ability. The AKM model exploits the connectedness of the director–firm matching network to separate persistent board member heterogeneity from persistent firm heterogeneity. This allows me to mute a fundamental –and typically unobserved– endogenous channel. The empirical design pairs these AKM controls with a comprehensive set of control variables, which are designed to control for all the known channels that make the relationship between interlocking directorates and innovation endogenous. The remaining variation is therefore intended to be as close as possible to selec-

tion on observables. As key robustness tests, I complement the baseline regressions with both (i) sensitivity analyses that quantify how strong residual selection on unobservables would have to be to overturn the coefficients of interest –specifically, these are the tests from [Oster \(2019\)](#), [Cinelli and Hazlett \(2020\)](#), and [Diegert, Masten and Poirier \(2025\)](#)–, and (ii) a set of event studies around the formation of a new interlock. I also analyze the external validity of the estimated board member fixed effects.

I find that interlocking directorates are associated with higher innovative output, measured as increases in both raw and citation-weighted counts of patents. At the same time, I find little evidence that these interlocks are associated with higher innovation inputs in the form of R&D levels and R&D intensity. I also find that, when two firms interlock with each other for the first time, they start cross-citing each other more in their patents. The sensitivity analyses indicate that unobservables would need to be much stronger than the rich set of included observables to overturn the headline estimates, indicating that omitted variable bias is an unlikely driver of these estimates.¹ For instance, the test from [Cinelli and Hazlett \(2020\)](#) indicates that one would need an unobservable covariate that is approximately 36.4 times stronger than firm size or 35.7 times stronger than R&D spending to nullify my estimate on the number of patents in the OLS regression.

To understand the mechanisms behind why interlocked firms patent more, I run a series of additional regressions and event studies to distinguish between strategic patenting behavior as opposed to technological learning. I find that, although interlocked firms patent more after treatment, these new patents are mostly continuation patents, which are commonly associated with strategic patenting and more litigation ([Berger, Blind and Thumm, 2012](#), [Hegde, Mowery and Graham, 2009](#), [Righi, 2026](#), [Righi and Simcoe, 2023](#), [Righi, Cannito and Vladasel, 2023](#)). Also, although they cite each other more often after treatment, these citations do not systematically expand into new partner patents or new partner technology classes, which is what one would expect if interlocks primarily opened new knowledge channels. Nor do the data show a clear increase in citations to technologically proximate partner patents. Instead, the stronger post-interlock responses are concentrated in citations to partner patents that are older and not technologically

¹The use of OLS regressions with a saturated vector of controls and sensitivity analyses in the absence of clear exogenous shocks is not new in empirical research in Economics. Arguably, the most famous example is [Altonji, Elder and Taber \(2005\)](#), in which the authors estimate the impact of Catholic high school attendance on college outcomes.

close to the citing patent. This would also explain why, in the baseline regressions, there is no statistically significant effect on the market-based values of these new patents: if they were truly innovative, we would expect to see a value-based response.

This interpretation carries some policy implications. If interlocks were mainly promoting knowledge diffusion, their effect on patenting could be viewed as at least partly innovation-enhancing. By contrast, the evidence here provides evidence that most of the post-interlock increase in patent activity reflects strategic portfolio management. No result presented in this paper suggests that board interlocks promote the production of higher-valued or more innovative patents. That matters because strategic patenting may raise legal congestion or increase the complexity of overlapping intellectual property claims (Shapiro, 2000). Therefore, the costs of interlocks may extend beyond product market coordination or information sharing in the usual antitrust sense. This may imply that regulators evaluating interlocks would want to pay close attention not only to product market effects but also to how board connections may alter innovation strategy, which in turn may affect market entry.

This paper contributes to three strands of literature. The first one is the literature on interlocking directorates. In the recent empirical branch of this literature, Poberejsky (2024) uses exogenous firm mergers to find that horizontal interlocks increase cumulative returns on assets. He also documents that, when a horizontal interlock breaks, patenting decreases.² Barone, Schivardi and Sette (2025) exploit the effect of a ban on interlocking directorates in the Italian banking industry to find that, as a result of the law, previously interlocked banks experienced a decrease in their interest rates compared to banks that were never interlocked. Herrera-Caicedo, Jeffers and Prager (2026) provide direct evidence that firms become substantially more likely to collude after they begin sharing executives or board directors. Whereas they study explicit coordination in labor markets, I examine whether the same organizational link reshapes firms' innovation behavior and the strategic use of intellectual property. Cabezón and Hoberg (2026) show that firms embedded in denser competitor board networks experience innovation herding, lower product differentiation, lower patent values, and poorer performance. Gualdani (2021)

²Apart from the empirical strategies being radically different, while Poberejsky (2024) looks at the effect of *losing* an interlocking directorate on financial returns using a difference-in-differences strategy, mine focuses on what happens when you *form* a new interlocking directorate. It is unlikely that the two phenomena have symmetric effects.

develops a game of network formation with an application to interlocking directorates and finds that a firm's payoff from having an executive sitting on the board of a competitor increases with the number of other competitors that have an executive appointed to that firm's board. Earlier qualitative and empirical work in sociology focused on the determinants of interlocking directorates and tried to uncover factors that are correlated with them (Dooley, 1969). These include studies on industry concentration and horizontal interlocks (Burt, 1983, Pennings, 1980), cooptation (Allen, 1974, Bunting, 1976, Palmer, Friedland and Singh, 1986, Pfeffer, 1972, Pfeffer and Salancik, 1978, Selznick, 1949, Stearns and Mizruchi, 1993, Thompson and McEwen, 1958), career advancement (Stokman, Knoop and Wasseur, 1988, Zajac, 1988), and social ties (Mills, 1956).

This paper contributes to this literature in several ways. First, it provides new firm- and pair-level evidence that interlocks reshape innovative activity not only on the intensive margin of patenting, but also in the direction of technological interaction between connected firms. Existing work has argued that interlocks may transmit information, facilitate coordination, and affect strategic behavior (Burt, 1983, Mizruchi, 1996, Pfeffer and Salancik, 1978). Nevertheless, we lack empirical evidence on how these channels affect firm innovation. I address this by analyzing patenting behavior, citations, and the timing of first interlock formation and showing that board connections are associated with a reorganization of firms' innovation activity. Second, the paper opens the black box of the mechanisms more directly than the existing literature by distinguishing between genuine knowledge transmission and defensive use of patents. At the firm level, I show that the increase in patenting is concentrated in continuation patents, which are commonly linked with strategic patenting motives. Using pair-level event studies, I then show that post-interlock citation responses are disproportionately concentrated in older, more prominent, and non-technologically proximate partner patents. Third, this paper offers a new organizational perspective to the literature on innovation by showing that the consequences of interlocking are not restricted to product market coordination but extend to how firms manage patenting portfolios and position themselves in the technological space. In that sense, the paper bridges the literatures on corporate governance, innovation, and intellectual property (Cohen, Nelson and Walsh, 2000, Hall and Ziedonis, 2001, Shapiro, 2000) by documenting that interlocks can operate through a strategic channel that has largely been overlooked in prior work.

The second strand of literature studies AKM models. This class of models was originally introduced by [Abowd, Kramarz and Margolis \(1999\)](#). Since its introduction, the AKM model has provided researchers with a way to quantify how much of wage inequality is attributable to worker-specific factors, firm-specific factors, and the sorting between high-wage workers and high-wage firms. In the recent past, this framework has been employed in several empirical applications (see [Card et al. \(2018\)](#) and [Kline \(2024\)](#) for a comprehensive overview), as well as sparking a lively literature on the estimation of the model. For instance, [Andrews et al. \(2008\)](#) showed that the so-called limited mobility bias can depress the estimated correlation between worker and firm effects, and they propose a bias-correction method.³ [Andrews et al. \(2012\)](#) report clear evidence that this bias masks positive assortative matching. [Kline, Saggio and Sølvesten \(2020\)](#) develop a leave-one-out correction for estimating the variance of firm effects and the covariance. The current consensus, accounting for the latest bias-correction methods, is that there is positive assortative matching in most labor markets. In the empirical application of this paper, however, I am not concerned about the estimation of such sorting. Instead, I use the AKM model only to estimate the individual fixed effects of the board members. In a correctly specified AKM model, worker and firm fixed effects are unbiased regardless of positive or negative sorting. Therefore, the correction approaches that the literature has discussed to correctly estimate sorting do not apply to my case.

To my knowledge, this is the first paper to use the AKM model in an intermediate step to control for an endogenous and unobservable channel. In my empirical strategy, I first estimate the AKM model for the largest connected sample of board members and firms. Then, I collect the estimated director fixed effects –which are, by definition, a proxy variable for their time-invariant ability– and aggregate them at the firm-year level to construct a measure of aggregate board ability for every company in each year. Using this firm-level aggregated AKM measure as a control variable allows me to mute the board member ability channel, which would otherwise complicate the analysis. In the corporate finance literature, [Bertrand and Schoar \(2003\)](#) exploit the movement of top executives

³Limited mobility bias can arise because worker and firm fixed effects in AKM models are identified from workers who move across firms. The connectedness of the worker–firm network is particularly weak when the mobility of workers across firms is sparse, which means that the sampling noise in estimated firm effects can attenuate the estimated covariance between worker and firm effects. In other words, if not too many workers move across firms in the data, it becomes difficult to distinguish whether high wages are due to the worker or the firm, which biases downward the estimates of assortative matching.

across firms to estimate manager fixed effects and show that managers account for persistent differences in corporate performance. [Graham, Li and Qiu \(2012\)](#) use the AKM model to study how firm and executive-specific effects explain executive compensation, and find that executive fixed effects explain a large portion of the variation in pay. [Cavaco et al. \(2017\)](#) applies the AKM model to find that having more independent directors on the board correlates with lower firm performance. Importantly, [Bertrand and Schoar \(2003\)](#) and [Graham, Li and Qiu \(2012\)](#) focus on top executives, while [Cavaco et al. \(2017\)](#) focuses on French board members, using independence as the key margin of interest. My AKM model differs from theirs both in the sample, as I look at the universe of executives and independent directors in public U.S. firms, and in the purpose, as I use the AKM fixed effect estimates as control variables in a second stage and not as an outcome of interest per se.

The third literature studies common ownership among rival firms and, more generally, the consequences of firms sharing the same resources.⁴ Work in this area analyzes how horizontal links created by overlapping shareholders reshape competition, managerial incentives, innovation, and merger incentives. [Azar \(2022\)](#) shows that firms with a high degree of common ownership are more likely to share directors, while [Azar, Schmalz and Tecu \(2018\)](#) argue that common ownership can soften product market competition and raise prices. [Antón et al. \(2023a\)](#) develop a mechanism through top management incentives and show that, when firms share important shareholders, managerial pay becomes less tied to aggressive relative performance. This in turn can dampen competitive pressure. Relatedly, [Antón et al. \(2023b\)](#) show that common ownership can affect merger incentives and acquisition activity, while [Ederer and Pellegrino \(2025\)](#) quantify the welfare costs of the interaction between ownership overlap and product market rivalry. [Eldar and Grennan \(2021\)](#) study common ownership in startups backed by venture capitalists and argue that it differs from passive institutional ownership of public firms, since venture capitalists are active owners and may create channels through which information and coordination pass across firms. This literature is relevant for my paper as it shows that, in general, horizontal links among rivals need not have a uniform effect on innovation. For instance, [Antón et al. \(2025\)](#) shows that the sign of this effect depends on whether the

⁴A related literature about mergers and innovation similarly emphasizes that horizontal integration can alter innovation incentives through the interaction of reduced competitive pressure and innovation externalities ([Federico, Langus and Valletti, 2017, 2018](#)).

dominant force is the internalization of technological spillovers or the softening of product market rivalry. Whereas common ownership links rivals through shared shareholders and changes incentives from above, interlocking directorates link rivals inside the boardroom and may operate through information transmission as well as through competition.

I contribute to this literature by showing that horizontal board links shape innovation not through broad increases in measured R&D, but through changes in the direction and strategic use of firms' patenting behavior. In particular, I show that while newly interlocked competitors increase cross citation activity, this response is concentrated in the partner patents that are older, more prominent, and non-technologically close to the citing patent, a pattern more consistent with strategic patenting than with technological learning. This paper complements the common ownership literature by identifying a board-level channel through which shared resources among competing firms can affect innovation and the management of patent boundaries, net of common ownership.

The rest of this paper is organized as follows. Section 2 illustrates the institutional context, the data I use, and a series of descriptive statistics. Section 3 lays out my identification strategy. Then, Section 4 shows the baseline results. After that, Section 5 discusses and shows results related to the mechanisms that may explain the results I document. Section 6 discusses the robustness of these results to alternative specifications. Finally, Section 7 concludes.

2 Institutional Context and Data

2.1 Section 8 of the Clayton Act and Recent Ramp-Up in Enforcement

Due to collusion concerns, interlocking directorates were deemed illegal in 1914 under Section 8 of the Clayton Antitrust Act and have remained so ever since. Not all forms of interlocking directorates are unlawful: only those between competing companies (i.e., in the same line of industry) that exceed size-based thresholds.⁵ Nevertheless, antitrust

⁵This revenue threshold is set at \$54,402,000 (for Section 8(a)(1) of the Clayton Act) as of the time of this writing. Interlocking directorates are still permissible for competing firms if and only if the sales of either corporation are less than \$5,440,200 (Section 8(a)(2)(A) of the Clayton Act), the competitive sales of either corporation constitute less than 2% of that corporation's total sales, or the competitive sales of both

enforcement against interlocking directorates has been notoriously lax since the establishment of this law. Between 1899 and 2022, the U.S. Department of Justice (DoJ) was involved in just eleven cases involving interlocking directorates ([Poberejsky, 2024](#)). Several reasons may explain this lenient antitrust stance, including for example ambiguity over the relevant notion of competition and the FTC’s broad discretion in exercising its enforcement powers ([Nili, 2022](#)).

With that being said, under the Biden administration in particular, there was a ramp-up in the enforcement of Section 8 of the Clayton Act. In 2021, the DoJ started renewing its scrutiny of interlocking directorates. In June of that year, they broke an interlock between Endeavor and Live Nation, two sports and entertainment companies that closely competed in several markets. Scrutiny of interlocking directorates only grew stronger after that, and in October 2022, seven more directors from five different companies also resigned in response to DOJ concerns about potentially illegal interlocks.⁶ In 2023, the FTC followed suit and announced an order that would prohibit Quantum from obtaining a seat on EQT’s Board of Directors following EQT’s acquisition of certain natural gas assets from private equity firm Quantum. Behind this decision was the FTC’s explicit worry that this board appointment would create a potentially illegal interlock.⁷

2.2 Data on Boards and Interlocking Directorates

The main source of information on boards in this paper is BoardEx, which assembles board rosters for U.S. public companies from disclosures filed with the Securities and Exchange Commission (SEC). For each appointment to a board, BoardEx records the names and the titles of the board members along with their demographic attributes, compensation components, and employment histories. From these microdata, I reconstruct board composition over time and identify interlocking directorates by identifying the firms that share at least one board member. To then classify horizontal ties, I implement a rule-based Section 8 screen. I first restrict attention to firm pairs that are competitors at the chosen

corporations constitute less than 4% of each corporation’s total sales. These thresholds are reviewed each year – usually in January – by the Federal Trade Commission (FTC).

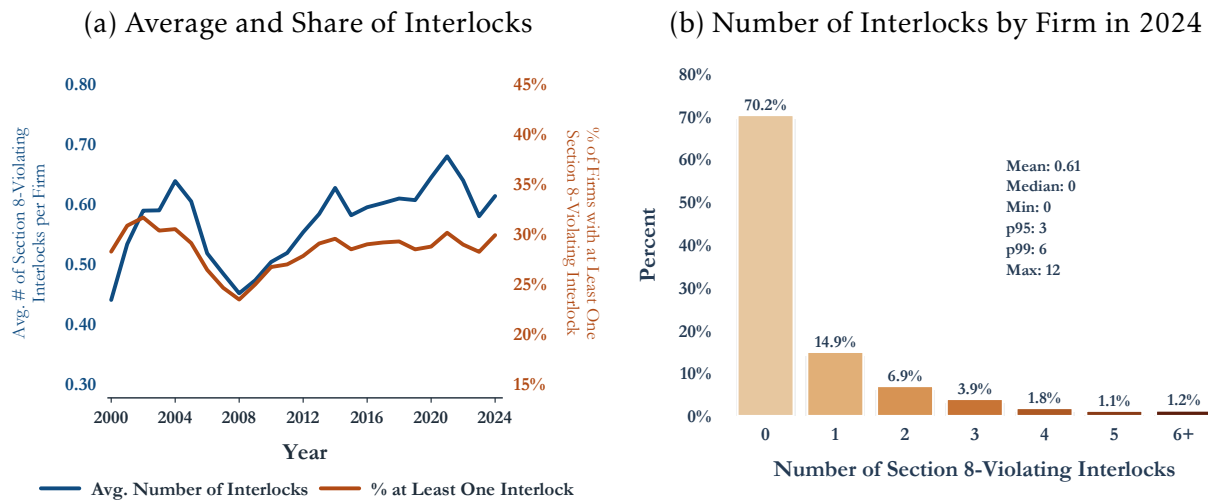
⁶[Directors Resign from the Boards of Five Companies in Response to Justice Department Concerns about Potentially Illegal Interlocking Directorates \(justice.gov\)](#)

⁷[FTC Approves Final Order to Prevent Interlocking Directorate Arrangement, Anticompetitive Information Exchange in EQT, Quantum Energy Deal \(ftc.gov\)](#)

industry cut⁸ and that share at least one board member (either an independent director or an executive) in a given year. I then merge the FTC’s annual Section 8 thresholds with this information and use Compustat to determine whether each firm is covered by the statute in a given year and is not exempt under absolute de minimis sales cutoff. Finally, I classify an observed horizontal interlock as Section 8–violating when both firms are eligible and neither belongs to an excluded entity type, such as banks or trust entities. Appendix A.1 provides the full algorithm and implementation details.

Figure 1 summarizes the prevalence of Section 8–violating horizontal interlocks in the sample. Panel 1a shows that illegal interlocks are far from rare: in recent years, roughly 30% of public firms in the U.S. engaged in the practice with a competitor, while the average number of interlocks per firm was 0.61 in 2024. Panel 1b shows the cross section from 2024 which highlights that, conditional on interlocking, most firms only had one horizontal interlock. Only a handful collects three or more at the same time.

Figure 1: Firm-Level Statistics About Horizontal Interlocking Directorates, 2000-2024



Notes: Panel (a) shows the average number of horizontal interlocking directorates (blue line, left axis) across all firms in the sample by year, and the percentage of firms that have at least one interlocking directorate by year (red line, right axis). Panel (b) shows the cross sectional distribution of the number of horizontal interlocking directorates by firm in 2024, along with the percentage of firms for each category.

Table 1 reports a series of statistics at the director-level, of which there are almost 100,000 in my sample. On average, a board member in the BoardEx panel spends 6 years

⁸I use NAICS-2 codes as a baseline but later relax this restriction to test for robustness.

at a firm, with independent directors enjoying slightly longer tenure on average than executives. Only 17.23% of the sample of board members in the data are women, with the proportion decreasing to around 5% for executive roles. The average director joins a new board around age 60 and departs at 65. Further, they hold almost two concurrent seats at any given time, and the overwhelming majority serve on at least one independent committee out of the traditional audit, compensation, and nomination committees.

Having illustrated the data I use for my analysis, I now report three empirical facts about interlocked directors, firms' likelihood to interlock, and the average persistence of interlocks.

Fact #1: Interlocked directors are more likely to be CEOs or Presidents. Figure 2 summarizes how interlocked directors differ from non-interlocked directors along a range of observable characteristics. Each point in the graph is an estimate that comes from a separate regression from the others, and it reflects both direct selection into interlocks and sorting across firms with different baseline propensities to interlock. I document that interlock status at the director-level is positively correlated with corporate authority. For instance, CEOs and Presidents are more likely to hold at least one concurrent board seat at another firm. The same graph also shows that interlocked directors have slightly higher compensation and larger professional networks. This is consistent with the idea that interlocked individuals are more productive, more in demand across firms, more influential, and therefore more likely to sit on the boards of multiple firms. Although descriptive, the pattern is nonetheless informative for my empirical strategy because it confirms that interlocks are not random assignments across individuals and disproportionately involve directors with greater managerial authority and broader networks, which are precisely the types of attributes that could otherwise confound inference.

Fact #2: Firms prefer to interlock with similar firms. Table 2 reports coefficients from logit regressions and characterizes interlock formation at the firm-pair level. The propensity to form an interlock between any two firms can be strongly predicted by proximity and similarity. For instance, geographic distance is negative and statistically significant across specifications, which implies that more distant dyads have lower odds of interlock formation (Azar, 2022). I find that being in the same industry also matters, because I illustrate that interlocks form disproportionately within sectors. Being close in the product market space further predicts interlock formation, as a high Hoberg–Phillips (Hoberg and

Phillips, 2010, 2016) similarity indicator enters positively and significantly in the baseline columns. In the more restrictive specifications that incorporate additional differences in board composition, the estimates continue to point toward assortative matching. Several differences enter with negative point estimates, including a statistically significant negative coefficient on differences in gender composition, which suggests that interlocks form more often between firms whose boards resemble each other. This is again consistent with a view of interlock formation as a non-random matching process in which firms connect to geographically proximate and compositionally or economically similar companies.

Table 1: Director-Level Summary Statistics, 2000-2024

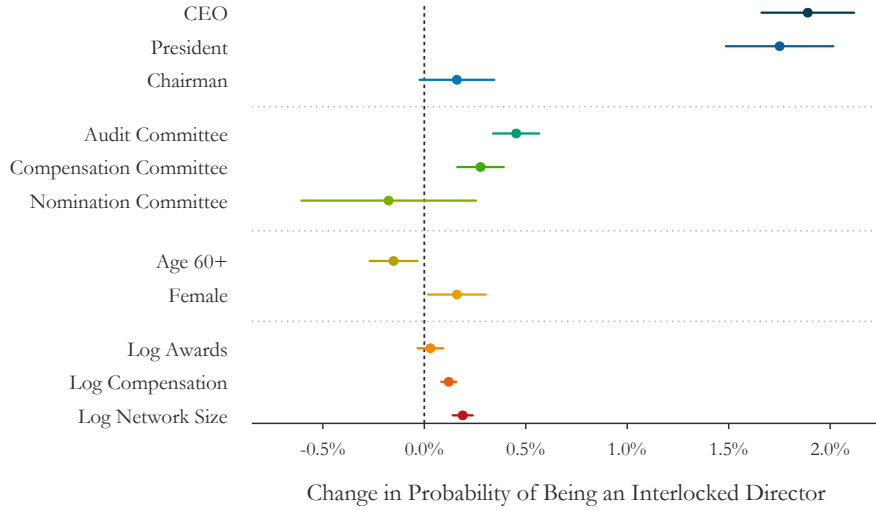
	(1) All	(2) Independent	(3) Executive
<i>Panel A: Director-Level Statistics</i>			
Tenure (years)	5.98	6.01	5.56
Age at Appointment	60.27	60.42	58.41
Age at Exit	65.25	65.43	62.97
Female (%)	17.23	18.19	5.05
<i>Panel B: Board Connectivity and Committees</i>			
Concurrent Seats (avg.)	1.80	1.82	1.44
Committee Roles (% any of A/C/N)	91.92	95.76	43.17
N (director–firm spells)	154,871	143,555	11,316
N (unique directors)	94,461	86,184	8,277

Notes: Columns in Panel A report means across director–firm spells (a unique director–board appointment), while columns in Panel B report means across the panel. “Independent” denotes non–executive directors, while “Executive” denotes inside directors. *Tenure (years)* is the length of the employment spell (last minus first appointment year +1). *Age at Appointment* and *Age at Exit* are computed at the first and last year of the appointment spell, respectively. *Female (%)* is the share of spells held by female directors. *Concurrent Seats (avg.)* is the mean number of simultaneous board seats the director holds in a given year within a spell. *Committee Roles (% any of A/C/N)* illustrates the share of spells in which the director served on at least one Audit, Compensation/Remuneration, or Nomination/Governance committee at any point during a spell. *N (director–firm spells)* counts unique spells; *N (unique directors)* counts distinct directors. All the variables are constructed from BoardEx.

Fact #3: Horizontal interlocks are typically persistent. Finally, in Figure 3, I take a look at the average length of a horizontal interlock. In particular, I report the observed spell length in years of this practice. I document that only around 15% of horizontal

interlocks dissolve in the first year, while around 50% last three or more years. Roughly 1 in 10 lasts at least eight years. Therefore, when two boards connect through an interlock, they usually do so for an extended period of time, either with just one interlocked director or several interlocked board members.

Figure 2: Director-Level Correlates of Being Interlocked



Notes: Each point is the OLS coefficient from a separate regression. Each regression uses an indicator for whether a director is interlocked as the outcome variable, and a single director-level characteristic as the explanatory variable. I use the total concurrent number of boards a director works for and year fixed effects as control variables. Formally, for each director d and characteristic k , the regression I estimate is $\mathbb{1}\{\text{Interlocked}_d\} = \alpha_k + \beta_k X_{d,k} + \gamma_t + \omega \log(\text{Number of Boards}) + \varepsilon_{d,k}$. The plotted β_k therefore summarizes the correlation between characteristic $X_{d,k}$ and interlocked status. The whiskers around the dots report the corresponding 95% confidence intervals.

2.3 Measures of Innovation

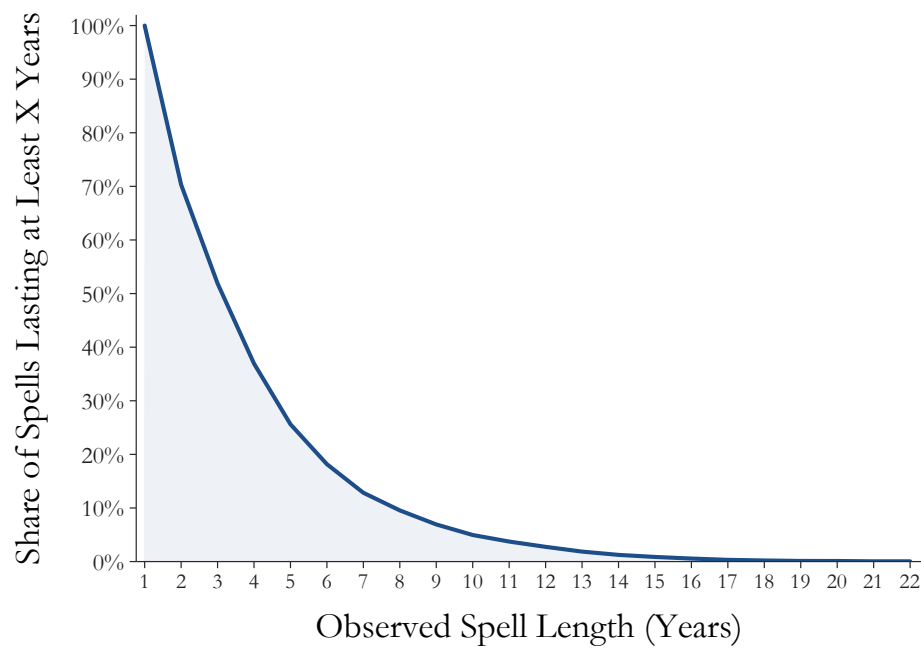
In my analysis, I distinguish between firm-level innovation inputs, outputs, and values. Inputs reflect the resources that firms devote to the innovation process, outputs capture the technological products of that process, and value measures the market-based economic importance of the innovations that are ultimately produced. In addition, I also study the capital-labor ratio as a complementary firm-level outcome, since interlocking

Table 2: Interlock Formation Between Firm Pairs

	(1) Prob(Interlock)	(2) Prob(Interlock)	(3) Prob(Interlock)	(4) Prob(Interlock)
Distance	-0.327*** (0.017)	-0.308*** (0.018)	-0.449*** (0.111)	-0.452*** (0.112)
Size Difference	-0.178*** (0.031)	-0.086*** (0.031)	-0.206** (0.104)	-0.029 (0.131)
R&D Difference	-0.225*** (0.022)	-0.213*** (0.022)	-0.064 (0.068)	-0.020 (0.070)
High Product Similarity	1.451*** (0.197)	1.250*** (0.186)	0.522 (1.203)	0.399 (1.107)
Same Industry		0.920*** (0.051)	0.572 (0.556)	0.546 (0.570)
Nationality Difference			-0.437 (0.454)	-0.465 (0.452)
Gender Difference			-2.012** (0.978)	-2.012** (0.967)
Busy Directors Difference			-0.504 (0.592)	-0.486 (0.594)
Network Size Difference			-0.000*** (0.000)	-0.000*** (0.000)
Time Spent on Board Difference			-0.026 (0.029)	-0.024 (0.029)
Revenue Difference				-0.016 (0.106)
COGS Difference				-0.225** (0.102)
Constant	-17.332*** (0.143)	-18.370*** (0.139)	-13.874*** (2.140)	-13.754*** (1.980)
N	2,205,283	2,205,283	151,408	151,401
Year FE	YES	YES	YES	YES

Notes: *, **, and *** denote $p < 0.10$, $p < 0.05$, and $p < 0.01$, respectively. Entries are logit coefficients, with standard errors in parentheses. All specifications include year fixed effects and two-way clustered standard errors. *High Product Similarity* is an indicator equal to one if the Hoberg–Phillips product-market similarity score for the firm pair falls in the top percentile of the corresponding year-specific distribution. All variables labeled *Difference* are constructed as absolute differences between the two firms in the dyad. *Size Difference* is the absolute difference in the number of employees. *Nationality Difference* is the absolute difference in the nationality composition of the two boards, as reported by BoardEx. *Gender Difference* is the absolute difference in the share of female directors across the two boards. *Busy Directors Difference* is the absolute difference in the share of busy directors across the two firms. *Network Size Difference* is the absolute difference in average director network size across the two boards. *Time Spent on Board Difference* is the absolute difference in average board tenure across the two firms, based on BoardEx records.

Figure 3: Spell Lengths of Horizontal Interlocking Directorates



Notes: This line graph reports the observed spell length (in years) for each horizontal interlocking directorate in the data, with the exclusion of those formed before the beginning of the panel. A spell length is defined as the difference between the year in which two firms first interlock and the year in which they dissolve that link. For each spell length, I report the percentage of interlocks that last for at least that period of time.

directorates may affect not only patenting behavior but also the organization of production and the relative use of capital and labor.

To measure innovation inputs, I use the R&D expenditure data in U.S. Dollars. I complement this with a measure of R&D intensity constructed as the ratio between R&D expenditures and total assets⁹. The above-mentioned capital-labor ratio concludes this first set of innovation measures, and is measured as the ratio between the net property, plant, and equipment (PP&E) figure and the number of employees, both again from Compustat. It is therefore thousands of dollars of net PP&E per employee. All the data used to construct these three measures are from Compustat.

To measure innovation outputs, I use two patent-based outcomes. The first is the raw count of new patents issued to a firm in a certain year. The second one is a citation-weighted count, defined as

$$\text{CW Patents}_{jt} = \sum_{p \in P_{jt}} \left(1 + \frac{C_p}{\bar{C}_p} \right),$$

where P_{jt} is the set of patents granted to firm j in year t , C_p is the number of forward citations received by patent p , and $\bar{C}_p$ is the average number of citations received by patents granted in the same year as patent p . Both the firm-level patent counts and the number of citations come from the dataset built by [Kogan et al. \(2017\)](#), which has now been extended to the end of 2024, and builds on the patent data from [Hall, Jaffe and Trajtenberg \(2001\)](#).

Finally, to measure the value of innovation, I again follow [Kogan et al. \(2017\)](#) and use stock market reactions around patent grants as a proxy for the private economic value of innovation ([Hall, Jaffe and Trajtenberg, 2005](#)). In particular, for each patent p , [Kogan et al. \(2017\)](#) estimate the firm's abnormal stock-market reaction, denoted by ξ_p , over the three-day event window around the grant date. I then aggregate these patent-level values to the firm-year level by summing across all patents granted to firm j in year t :

$$\text{Total Value}_{jt} = \sum_{p \in P_{jt}} \xi_p,$$

⁹In Compustat, R&D spending is missing for many firms. Therefore, I follow the standard in the literature and replace missing R&D values with zeros and then use a dummy variable equal to one if the reported R&D is zero as a control variable ([Antón et al., 2025](#)).

Table 3: Mean Firm-Level Summary Statistics for All Firms and All Available Years

	(1) All Firms (2000–2024)
<i>Panel A: Firm-Level Statistics</i>	
Assets	14,173.82
Revenue	6,302.94
COGS	4,159.07
Employees (k)	17.16
Patenting Dummy	0.38
<i>Panel B: Outcome Variables</i>	
R&D	169.43
R&D Intensity	0.03
Patents	26.24
Citation-Weighted Patents	25.69
Total Patent Value	494.42
Average Patent Value	8.39
Capital-Labor Ratio	402.01
N (firm-years)	10,034
N (firms)	1,949

Notes: Columns report means across firm-years. Assets, Revenue, COGS, and R&D are in U.S. dollars (millions). Employees are in thousands. R&D Intensity is the ratio of R&D to assets. Patenting Dummy is a dummy variable equal to 1 if a firm patented at least once during the year.

where P_{jt} again refers to the number of patents granted to firm j in year t . This variable captures the total private economic value that the stock market attributes to the firm's newly granted patents in that year. I also construct an average-value measure,

$$\text{Average Value}_{jt} = \frac{\text{Total Value}_{jt}}{P_{jt}},$$

which captures the mean private economic value of the patents granted to firm j in year t . Table 3 summarizes the variables introduced above.

3 Empirical Strategy

Let j index firms and t years. Use Y_{jt} to denote an outcome of interest (e.g., patent counts, citation-weighted patents, log R&D, or patent-value measures). Let Interlock_{jt} denote the interlock treatment, measured as an indicator for whether firm j holds at least one relevant interlocking directorate in year t . The baseline estimating equation is

$$Y_{jt} = \beta \text{Interlock}_{jt} + X_{jt}\phi + A_{jt}\pi + \psi_j + \lambda_{s(j)t} + \eta_{r(j)t} + \varepsilon_{jt}, \quad (1)$$

where ψ_j are company fixed effects, $\lambda_{s(j)t}$ are industry×year fixed effects (with $s(j)$ denoting the firm's industry), and $\eta_{r(j)t}$ are state×year fixed effects (with $r(j)$ denoting the firm's headquarters U.S. state). The vector X_{jt} collects a rich set of time-varying firm and governance controls, chosen using double LASSO. These controls come from Compustat and BoardEx. The coefficient of interest, β , captures the association between interlocks and the outcome. In an alternative specification,

$$Y_{jt} = \beta \text{Interlock}_{jt} + X_{jt}\phi + A_{jt}\pi + \psi_j + \tau_t + \varepsilon_{jt}, \quad (2)$$

one may use company fixed effects ψ_j and year fixed effects, τ_t . Each table will report estimates from both Equation 1 and 2.

These equations also include A_{jt} , which denotes board ability, which means that the neglect of A_{jt} from the estimated regression would lead to omitted variable bias. Ability is something we usually cannot explicitly control for, as it is unobserved in most empirical

settings. This is why, in this paper, I introduce a novel identification strategy that hinges on the AKM model to partial out the effect of A_{jt} on Y_{jt} . In particular, I first estimate board member ability in an intermediate step using the AKM model. This is a framework that originates from [Abowd, Kramarz and Margolis \(1999\)](#), which they used to decompose wage variation between worker and firm effects. In its canonical form, the AKM is a two-way fixed effect model:

$$W_{ijt} = \gamma_i + \psi_j + X_{jt} \theta + Z_{it} \xi + \epsilon_{ijt}, \quad (3)$$

where W_{ijt} is the wage earned by worker i in firm j at time t . This wage is then regressed on both worker fixed effects and firm fixed effects, along with a vector of control variables (X_{jt}, Z_{it}) collecting firm- and director-level observables. In the presence of the standard assumptions of this model, namely, mean independence of the error term and no match effects, the worker and firm fixed effects are unbiased. In my application on board members, I use the total compensation package (which includes bonuses and options on top of the baseline salary) as my measure of W_{ijt} . I collect this compensation data from ExecuComp.

There is a large and rich literature that deals with bias in the measurement of sorting and variance decomposition in this model ([Andrews et al., 2008, 2012](#), [Bonhomme et al., 2023](#), [Card et al., 2018](#), [Card, Heining and Kline, 2013](#), [Eeckhout and Kircher, 2011](#), [Kline, Saggio and Sølvsten, 2020](#)). However, in my empirical framework, I do not compute plug-in variance decompositions. My exclusive purpose is to use the AKM model as illustrated in Equation 3 to recover $\hat{\gamma}_i$ as a proxy for director ability. After estimating the model, I collect the $\hat{\gamma}_i$ and use them to construct board-level ability aggregates at the firm-year level and obtain A_{jt} . These firm-level AKM aggregates enter downstream regressions as controls for the ability channel that simultaneously drives interlock formation and outcomes. I restrict these further analyses to firm-years in which the full board is observed. The individual fixed effects are known to always be unbiased and consistent in the presence of complete information about the workers' wages ([Kline, 2024](#)).

Even after controlling for A_{jt} and all other channels, one cannot rule out remaining selection on unobservables using OLS. We can, however, determine how large unobservables would have to be to nullify the estimates on β from Equations 1 and 2. Hence why, in the second part of my identification strategy, I complement the AKM and baseline regressions with a range of modern sensitivity analyses ([Altonji, Elder and Taber, 2005](#), [Cinelli](#)

and Hazlett, 2020, Diegert, Masten and Poirier, 2025, Masten and Poirier, 2025, Oster, 2019). Every regression that I estimate in this paper is complemented by several tests: the Oster- β and Oster- δ estimates from Oster (2019), the breakdown sensitivity point from Diegert, Masten and Poirier (2025), and the sensitivity analyses from Cinelli and Hazlett (2020). Each time, I explain how to interpret these estimates, while Appendix A.2 contains an in-depth mathematical illustration of each of these tests.

4 Results

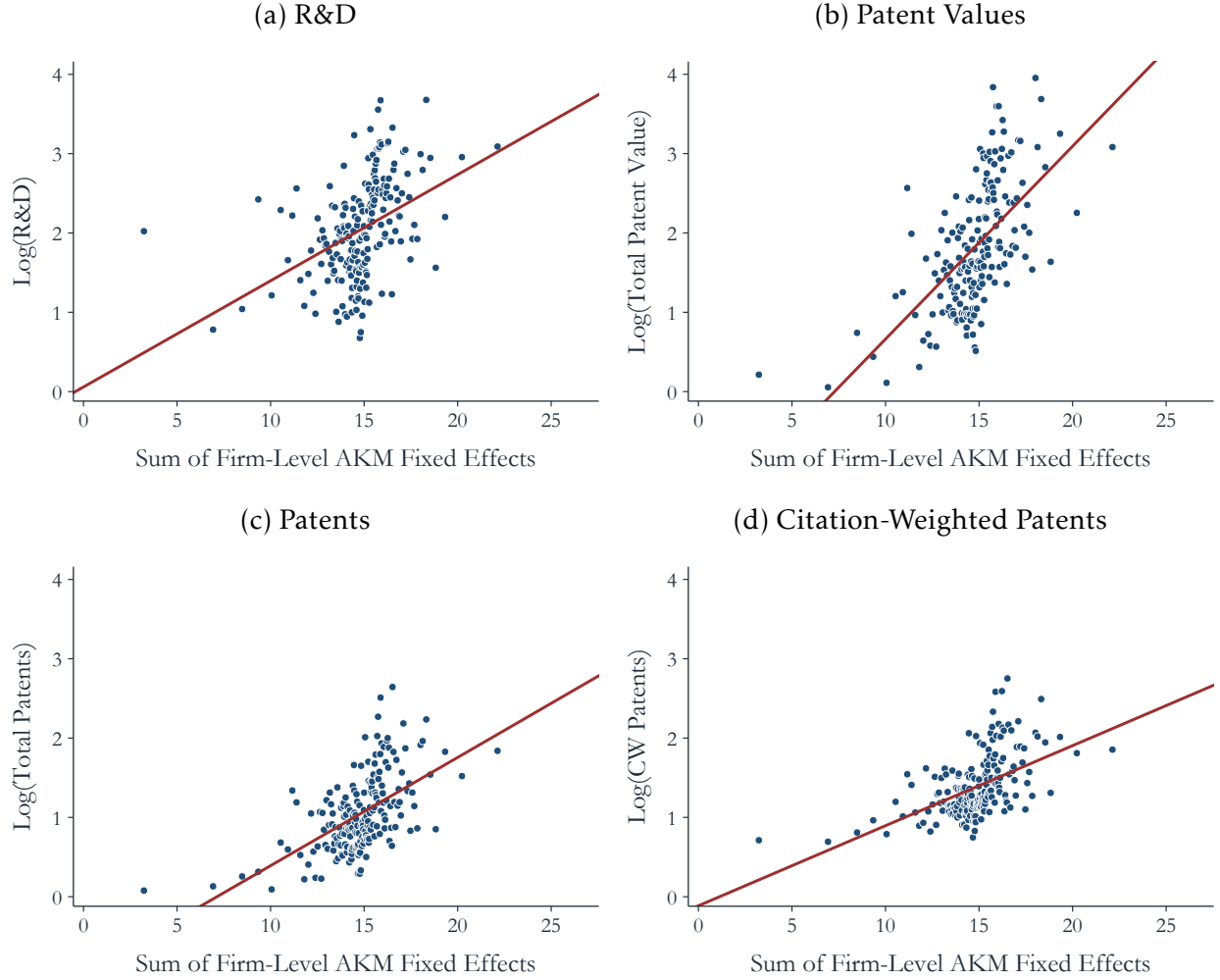
4.1 AKM Model

First, I investigate whether the firm-level AKM aggregates are systematically related to innovation outcomes. If the AKM director fixed effects capture a relevant dimension of board member ability, then firms with boards composed of directors with higher estimated fixed effects should also exhibit stronger innovative performance. Or, equivalently, better board members should select into better, more innovative firms. To evaluate whether this pattern holds in the data, I plot a set of binscatter relationships between the firm-level AKM ability measures and several innovation outcomes. Figure 4 shows that the relationship is indeed strongly positive throughout: firms with higher estimated board ability spend more on R&D, produce more patents, and generate patents that are both more highly cited and more valuable in market terms.

Further, the director fixed effects from the AKM model reflect the time-invariant ability of each board member. They are, effectively, the residuals of their compensation after partialling out time trends, company fixed effects, individual and firm characteristics, and other relevant covariates that may affect how much they get compensated. If better board members enjoy better careers and sort into more successful firms, as indeed several papers show (Baltrunaite and Karmaziene, 2024, Ferris, Jagannathan and Pritchard, 2003, Masulis and Mobbs, 2014, Yermack, 2004), then simple regressions should show that directors with higher estimated AKM fixed effects should, for example, work for larger, productive firms and have more seats throughout their career.

Table 4 tests precisely this hypothesis. For easier interpretation, in this set of regres-

Figure 4: Correlations Between Firm-Level AKM Director Estimates and Outcomes



Notes: These binscatter plots show the correlation between firm-year aggregates of estimated board-member fixed effects and firm-year outcomes. The horizontal axis is partitioned into 100 quantile-spaced bins, and each blue dot reports the mean of the x-variable and the mean of the y-variable within a bin. The red line reports the corresponding linear fit.

sions, I standardize the director fixed effects first. This way, each coefficient reflects a change in one standard deviation of the director’s ability on the outcome. In short, the estimated director ability is a strong predictor of career success. An increase in one standard deviation of the director’s ability is associated with, on average, 2.3% more simultaneous board seats. Better directors are also much more likely to work for larger firms, companies with bigger revenue, and firms that are in the S&P 500 and/or the Fortune 500. Finally, they enjoy slightly longer careers, too.

Table 4: Director Career Outcomes and the AKM Director Fixed Effect

	Log(Seats)	Log(Firm Size)	Log(Revenue)	Log(1+S&P 500)	Log(1+Fortune 500)	Log(Career Length)
AKM Director FE (SD)	0.023*** (0.004)	0.230*** (0.015)	0.201*** (0.014)	0.057*** (0.003)	0.044*** (0.003)	0.016*** (0.002)
N	16,389	16,318	16,318	16,389	16,389	16,389
R-squared	0.222	0.052	0.049	0.046	0.070	0.399

Notes: *, **, and *** denote $p < 0.10$, $p < 0.05$, and $p < 0.01$, respectively. Robust standard errors are reported in parentheses. The table reports OLS estimates from six separate cross-sectional regressions. The unit of observation is a director. The sample is US directors for whom a director fixed effect is estimated from the AKM model, standardised to mean zero and unit variance over the estimation sample. Coefficients illustrate the change in the outcome associated with a one standard deviation increase in the estimated director fixed effect. All specifications control for gender, decade of birth indicators, indicators for the decade in which they were first active, and years active. Years active is omitted from the career-length regression where it is collinear with the outcome. *Seats* is the number of simultaneously held US board seats averaged over active years. *Firm Size* and *Revenue* are total assets and revenue from Compustat and are averaged over director–firm–year service years. *S&P 500* and *Fortune 500* are counts of appointments at firms holding that status in the appointment year. *Career Length* is the span in years from a director’s first to last recorded role.

Throughout the years, several methods have been developed to test whether the assumptions of the canonical AKM model hold. I discuss these tests in Appendix A.3. There I show, for example, that the additively separable and exogenous mobility assumptions are satisfied in my data.

4.2 Baseline Regressions

Tables 5–7 report baseline firm-year estimates of Equation 1 and 2 for horizontal interlocks. The treatment variable, $Interlock_{jt}$, is an indicator equal to one if a firm j has at least one horizontal interlocking directorate in year t , and zero otherwise. All specifications condition on a rich set of firm controls chosen by double LASSO to control for observed

channels and absorb company, industry×year, and state×year fixed effects.¹⁰ Further, all these specifications always include the aggregated firm-level AKM estimates discussed in the previous subsection to partial out board member ability, and covariates that control for technological and product market spillovers.¹¹

Table 5 focuses on innovation inputs and production structure. The estimated association between horizontal interlocks and R&D inputs is economically small and statistically indistinguishable from zero. The point estimates for both R&D and R&D intensity are essentially zero. In contrast, the production structure outcome exhibits a modest positive association: the coefficient is 0.051 or 0.048 and is statistically significant at the 5% or 10% level, depending on the exact specification. Therefore, conditional on the full set of covariates, interlocks do not map into higher measured R&D investment but are associated with a slightly higher capital–labor ratio. The corresponding R^2 values for these three outcomes are quite close to 1, especially for the R&D and the capital-labor ratio regressions: this is desired in this particular empirical strategy, as I am trying to partial out as many channels as possible.

Table 5: Effects of Horizontal Interlocks on Innovation Inputs and Production Structure

	(1) Log(R&D)	(2) Log(R&D)	(3) Log(R&D Int.)	(4) Log(R&D Int.)	(5) Log(Cap/Lab)	(6) Log(Cap/Lab)
Interlock	-0.009 (0.028)	-0.004 (0.024)	-0.004 (0.003)	-0.002 (0.003)	0.051** (0.026)	0.048* (0.025)
N	4,210	4,451	4,257	4,497	3,354	3,586
R-squared	0.990	0.987	0.860	0.845	0.972	0.964
Firm FE	YES	YES	YES	YES	YES	YES
Year FE	NO	YES	NO	YES	NO	YES
Industry-Year FE	YES	NO	YES	NO	YES	NO
State-Year FE	YES	NO	YES	NO	YES	NO

Notes: *, **, and *** denote $p < 0.10$, $p < 0.05$, and $p < 0.01$, respectively. Standard errors (clustered by firm) are reported in parentheses. The unit of observation is the firm-year. *Interlock* is an indicator equal to one if the firm has at least one Section 8–violating horizontal interlock in year t and zero otherwise. All specifications include a set of firm-level controls chosen by double LASSO, the estimated firm-level AKM effects, and the corresponding sets of fixed effects as reported in the table.

¹⁰In Appendix A.6, I report an additional outcome variable which is outside of the scope of this paper but may nevertheless be of interest to the reader, which is firm-level markups. These markup measures are constructed using the De Loecker and Warzynski (2012) method. In short, I do not find statistically significant results for this outcome.

¹¹In Appendix A.7, I report heterogeneity results by industry.

Table 6 turns to innovation outputs. Here, we can see that horizontal interlocks are positively associated with patenting, both in raw counts and when weighted by citations. The estimated coefficients range from 0.081 to 0.115 log points for the raw patent count and from 0.072 to 0.087 log points for the citation-weighted equivalent, implying roughly a 8–12% increase in raw patent output and a 7–9% increase in citation-weighted output, depending on the specification. Relative to the input results, the output estimates are both larger and more precisely estimated.

Table 6: Effects of Horizontal Interlocks on Innovation Outputs

	(1) Log(Patents)	(2) Log(Patents)	(3) Log(CW Patents)	(4) Log(CW Patents)
Interlock	0.081* (0.043)	0.115*** (0.043)	0.072* (0.040)	0.087** (0.041)
N	3,889	4,121	3,933	4,165
R-squared	0.936	0.920	0.898	0.871
Firm FE	YES	YES	YES	YES
Year FE	NO	YES	NO	YES
Industry-Year FE	YES	NO	YES	NO
State-Year FE	YES	NO	YES	NO

Notes: *, **, and *** denote $p < 0.10$, $p < 0.05$, and $p < 0.01$, respectively. Standard errors (clustered by firm) are reported in parentheses. The unit of observation is the firm-year. *Interlock* is an indicator equal to one if the firm has at least one Section 8–violating horizontal interlock in year t and zero otherwise. All specifications include a set of firm-level controls chosen by double LASSO, the estimated firm-level AKM effects, and the corresponding sets of fixed effects as reported in the table.

Finally, Table 7 examines innovation values. The point estimates for the total value of patents are 0.021 or 0.059 log points for interlocked firms depending on the specification, while the average value estimates are negative, but none of these coefficients are statistically distinguishable from zero at conventional levels. Note that, if the number of total patents increases while their average value stays constant, the total value should mechanically increase. Indeed, the estimated coefficients for the total values are positive even though the average value decreases on average, although the correspondingly large standard errors prevent the establishment of any meaningful statistical result about them.

Table 7: Effects of Horizontal Interlocks on Innovation Values

	(1) Log(Total Value)	(2) Log(Total Value)	(3) Log(Avg. Value)	(4) Log(Avg. Value)
Interlock	0.021 (0.075)	0.059 (0.070)	-0.020 (0.057)	-0.036 (0.049)
N	3,704	3,930	3,237	3,470
R-squared	0.925	0.899	0.853	0.791
Firm FE	YES	YES	YES	YES
Year FE	NO	YES	NO	YES
Industry-Year FE	YES	NO	YES	NO
State-Year FE	YES	NO	YES	NO

Notes: *, **, and *** denote $p < 0.10$, $p < 0.05$, and $p < 0.01$, respectively. Standard errors (clustered by firm) are reported in parentheses. The unit of observation is the firm-year. *Interlock* is an indicator equal to one if the firm has at least one Section 8-violating horizontal interlock in year t and zero otherwise. All specifications include a set of firm-level controls chosen by double LASSO, the estimated firm-level AKM effects, and the corresponding sets of fixed effects as reported in the table.

4.3 Sensitivity Analyses

Even after controlling for director ability and all observable channels, one may still be worried about remaining selection on unobservables using OLS as an empirical framework. Nevertheless, we can use sensitivity analysis tests (Altonji, Elder and Taber, 2005, Cinelli and Hazlett, 2020, Diegert, Masten and Poirier, 2025, Masten and Poirier, 2025, Oster, 2019) to quantify how strong the remaining selection on unobservables would have to be to nullify our coefficients of interest. In this Section, I report the sensitivity analyses for the innovation outputs, since these are the outcomes that are statistically significant. In the Appendix, I further report the same analyses for innovation inputs and values, also. Finally, for brevity, in the main text I will explain these analyses mostly in words. For an in-depth review of the mathematics behind these tests, I refer the reader to Appendix A.2.

Table 8 shows the statistics from the state-of-the-art sensitivity analysis test from Cinelli and Hazlett (2020) for the innovation output coefficients. Column (1) reports the same point estimate from Table 6, while columns (2) and (3) report its associated homoskedastic standard errors and t-statistics. Column (4) is the share of the residual variance of the outcome, after the controls and fixed effects are partialled out, that in-

terlocks explain. The small values are consistent with the saturated design, as the fixed effects absorb the majority of the variance. Column (5) is the robustness value, or the minimum strength a confounder would need to have with both the treatment and the outcome, in equal measure, to drive the corresponding estimate to zero. For example, the first row tells us that an unobservable would have to explain 4.62% of the residual variance of the interlock status and also 4.62% of the residual variance of log patents –after partialling out all the covariates and fixed effects– to drive the coefficient to zero.

Is this a small or large number? [Cinelli and Hazlett \(2020\)](#) give no universal threshold, and the number is meaningless without a benchmark. Their advice is to benchmark this estimate against strong observed confounders. This is where the comparison with columns (7), (8), (9), and (10) is useful. Column (7) reports the share of the residual variance of the outcome that would be explained by an unobserved confounder assumed to be exactly as strong as firm size, measured by total assets, and column (8) reports the share of the residual variance of the treatment that the same confounder would explain. Columns (9) and (10) report the corresponding estimates for a confounder as strong as R&D. These results tell us that driving the estimate to zero requires an unobservable covariate explaining at least 4.62% of the residual variance of both the outcome and the treatment status. But a confounder as influential as total assets would explain 0.06% of the residual variance in patenting and 0.28% of that in interlock status, which is roughly 1.30% or 6.06% of what is required on each margin. These imply a Cinelli–Hazlett bias factor equal to only about 2.7% of that required to eliminate the estimated coefficient. In other words, one would require an unobservable some 36.4 times stronger than firm size to overturn the results for patent counts. Similarly, a confounder as influential as R&D spending would explain 0.88% of the residual variance in patenting and 0.02% of that in interlock status. Because bias requires association with both, such a confounder accounts for roughly 2.80% of the estimated coefficient according to the [Cinelli and Hazlett \(2020\)](#) bias expression, and confounding some 35.7 times stronger than R&D spending would be needed to overturn the result. Confounding on that scale, operating within firm, industry-by-year and state-by-year variation, is not something the observed covariates give any reason to expect. Very similar observations apply to the other three rows. In Appendix [A.2](#), I also report the results using the sensitivity analysis tests from [Oster \(2019\)](#) and [Diegert, Masten and Poirier \(2025\)](#), which confirm that these estimates are robust to omitted variable bias.

Table 8: Cinelli-Hazlett Sensitivity Analyses on the Effects of Horizontal Interlocks on Innovation Outputs

Outcome	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	Estimate	Std. err.	<i>t</i> -value	Treatment: Interlock $R^2_{Y \sim D X}$ (%)	RV (%)	df	Bound: LOG(AT) $R^2_{Y \sim Z D,X}$ (%)	Bound: LOG(AT) $R^2_{D \sim Z X}$ (%)	Bound: LOG(XRD) $R^2_{Y \sim Z D,X}$ (%)	Bound: LOG(XRD) $R^2_{D \sim Z X}$ (%)
Log(Patents)	0.08	0.03	2.44	0.22	4.62	2660	0.06	0.28	0.88	0.02
Log(Patents), Firm + Year FE	0.12	0.03	4.04	0.46	6.57	3523	0.02	0.07	0.17	0.01
Log(CW Patents)	0.07	0.04	1.98	0.15	3.74	2696	0.06	0.45	0.18	0.02
Log(CW Patents), Firm + Year FE	0.09	0.03	2.78	0.22	4.54	3561	0.01	0.14	0.04	0.01

Notes: Each row reports the sensitivity of the Interlock coefficient in the corresponding OLS specification to unobserved confounding, following the framework of Cinelli and Hazlett (2020). Estimate, Std. err., *t*-value and df are obtained by re-estimating that specification by OLS with its own fixed effects entered as dummy variables, on the same estimation sample as the corresponding baseline regression. The point estimates coincide with those reported in the baseline OLS tables, but the standard errors are homoskedastic and are *not* the firm-clustered standard errors reported there. $R^2_{Y \sim D|X}$ is the partial R^2 of the treatment with the outcome. RV is the robustness value: the minimum strength of association, as a share of residual variance, that an unobserved confounder would need with both the treatment and the outcome to reduce the estimate to zero. The last four columns bound the strength of a confounder Z assumed to be as strong as the named observed covariate ($k_D = k_Y = 1$). The bounds report: $R^2_{Y \sim Z|D,X}$ is its share of the residual variance of the outcome and $R^2_{D \sim Z|X}$ its share of the residual variance of the treatment. A benchmark whose implied values fall below RV cannot, on its own, overturn the estimated effect.

4.4 Event Studies

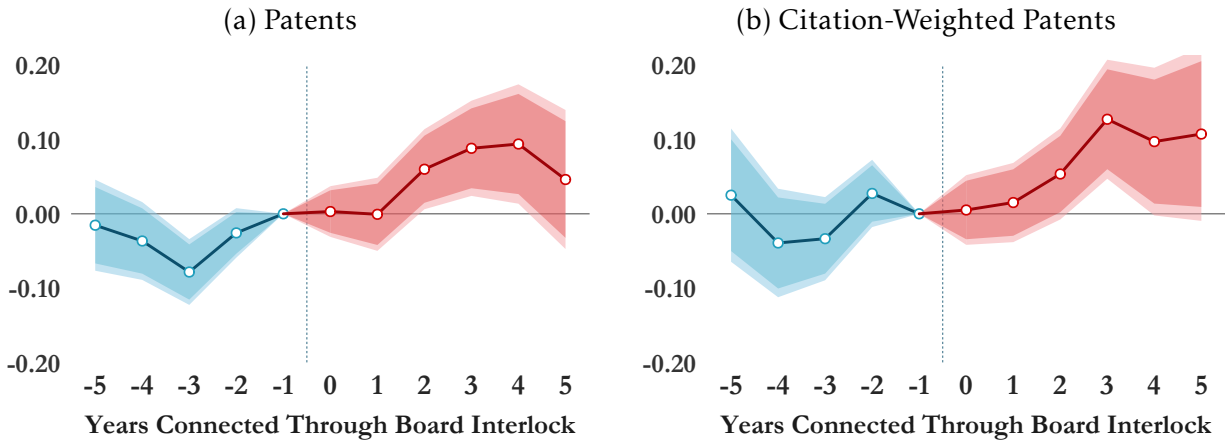
As a further robustness check, we can also inspect patent outcomes around the time when two firms interlock with each other for the first time. If interlocking directorates do indeed spark an increase in innovation, we should be able to see a larger increase in the number of total patents issued by the two interlocking firms compared to a control group formed by non-interlocking dyads. In particular, I estimate a dyad-level event study using the staggered adoption difference-in-differences estimator from [Callaway and Sant’Anna \(2021\)](#). Let G_{fj} denote the first year in which dyad (f, j) becomes interlocked. For each treatment cohort g and calendar year t , the estimator estimates a group-time average treatment effect, $ATT(g, t)$, comparing dyads first treated in g with never-treated dyads. I then aggregate these $ATT(g, t)$ ’s by event time $k = t - g$:

$$\theta_k = \sum_{g: g+k \leq T} w_{g,k} ATT(g, g+k), \quad (4)$$

with $k = -1$ normalized to zero. $w_{g,k} = \Pr(G = g \mid G + k \leq T)$ is the relative share of treatment cohort g among cohorts observed at event time k .

The results are shown in Figure 5 for the patent counts, both raw and weighted by citations. In particular, 5a shows the total number of patents issued by the dyads and 5b shows the total number of citation-weighted patents, again at the dyad level. In both cases, patent counts increase in the medium-to-long run following the establishment of a new interlock between two firms. Specifically, there is a non-statistically significant effect in the two years after the interlocking event, but then the count of patents starts increasing from the third year and becomes statistically significant, which signals a slow, steady increase in innovation following a new interlock rather than a rapid jolt immediately following the event. This is also consistent with the view that innovation and production take some time to adjust following an event that may impact them. Further, note that the magnitudes of the coefficients in these event studies match those from the OLS regressions: Figure 5 shows that, a few years after the formation of a new interlocking directorate, interlocked firms issue on average seven to thirteen percent more patents than non-interlocked firms, which is similar to the range reported in Table 6. Despite these analyses being fundamentally quite different, they lead to the estimation of the same sign and similar range of magnitudes.

Figure 5: Event Studies - Patent Counts



Notes: Each panel reports staggered-adoption event study estimates of the effect of first adopting an interlock on the stated outcome. The reported estimates are from the [Callaway and Sant'Anna \(2021\)](#) framework, which maps cohort- and time-specific $ATT(g, t)$ to event time $k = t - g$, with $k = -1$ normalized to zero. Event time is defined relative to the first interlock ($k = 0$) between any two firms. The dots are the estimates from such regressions. The darker red band shows the 90% confidence intervals, whereas the lighter red band shows the 95% confidence intervals. The standard errors are clustered at the firm-pair level. Logged outcomes use $\log(1+x)$. All specifications include calendar-year fixed effects and the baseline dyad-level controls as described in the text. *Patents* and *Citation-Weighted Patents* are defined at the dyad level.

5 Mechanisms

5.1 Patent Citations

Why does innovation increase after the formation of an interlock? The question may not only be interesting per se, but may also carry policy implications. For example, an increase in patents may be consistent with a parallel increase in the transmission of know-how between the two competitors (Stein, 2008). This may lead either side of the dyad to innovate and formulate new patents that may otherwise never see the light of day. If that is the case, the lax approach on the part of U.S. authorities may have been beneficial to American innovation over the years. On the other hand, the same increase in patents could also be explained by a more strategic or defensive approach. According to this alternative theory, firms use their board links to learn about the interlocked partners' innovation portfolios and use that to issue patents that may guard their own innovation.

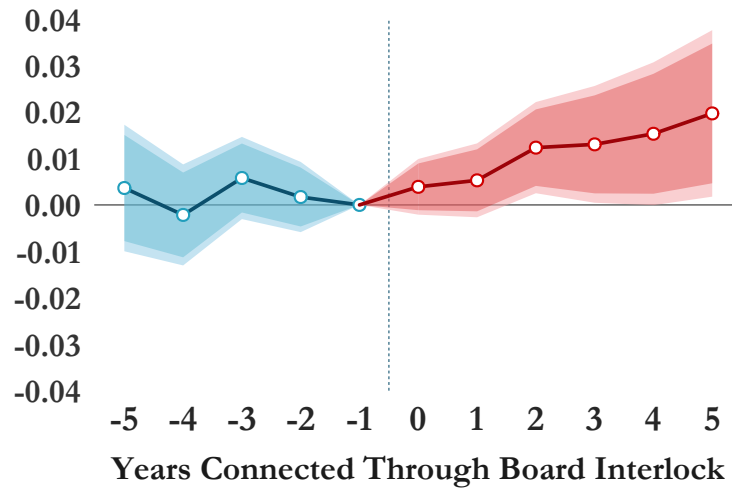
In this section, I probe these mechanisms. In Subsection 5.1.1, I explore whether the increase in patents after an interlocking event is mirrored by an increase in cross-citations between the two firms. This will give us a sense of whether the new interlocked firms show greater attention to the partner's patents to begin with. In Subsection 5.1.2, I explore whether these dynamics differ between origin and destination firms. Finally, in Subsection 5.2, I study which patents get issued after the formation of an interlock, including a look at continuing patents. If this business practice is consistent with a theory of positive know-how use, we should see interlocking firms making more innovative patents. For example, we should expect that a newly interlocked firm starts innovating in areas or patent classes that the interlocked partner is already familiar with. On the other hand, if a more strategic channel dominates, firms should be less likely to enter the partner's technological space and more likely to cite the partner's prior patents in ways that are more consistent with protecting their existing product market positions.

5.1.1 Cross-citations and citations per patent

All the event studies in this and subsequent subsections follow the same dyad-level design illustrated in Equation 4. To begin, Figure 6 reports the event-study dynamics for total cross citations between two firms. We can see that, after the first interlock, cross citations

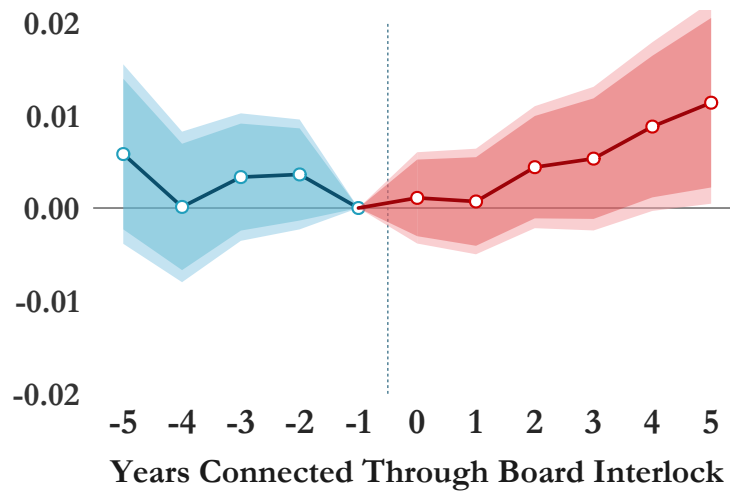
rise steadily and become statistically different from zero, reaching roughly 2 percentage points in magnitude by the end of the timeframe. Figure 7 complements these findings by looking at an extensive margin component, i.e., whether an interlock increases the probability that the pair exchanged any citation link in a given year. Again, we see that the post-treatment effects steadily grow over time. Therefore, interlocks open new citation relationships at the patent level rather than just amplifying citation intensity within an already-citing subset of pairs.

Figure 6: Event Studies - Intensive Margin of Patent Citations



Notes: This figure reports staggered-adoption event study estimates of the effect of first adopting an interlock on the stated outcome. The reported estimates are from the [Callaway and Sant'Anna \(2021\)](#) framework, which maps cohort- and time-specific $ATT(g, t)$ to event time $k = t - g$, with $k = -1$ normalized to zero. Event time is defined relative to the first interlock ($k = 0$) between any two firms. The dots are the estimates from such regressions. The darker red band shows the 90% confidence intervals, whereas the lighter red band shows the 95% confidence intervals. The standard errors are clustered at the firm-pair level. Logged outcomes use $\log(1 + x)$. All specifications include calendar-year fixed effects and the baseline dyad-level controls as described in the text. *Cross-Citations* is the share of a firm-year's patent citations that go to (or come from) firms with which the focal firm shares at least one director in year t .

Figure 7: Event Studies – Extensive Margin of Patent Citations



Notes: This figure reports staggered-adoption event study estimates of the effect of first adopting an interlock on the stated outcome. The reported estimates are from the [Callaway and Sant'Anna \(2021\)](#) framework, which maps cohort- and time-specific $ATT(g, t)$ to event time $k = t - g$, with $k = -1$ normalized to zero. Event time is defined relative to the first interlock ($k = 0$) between any two firms. The dots are the estimates from such regressions. The darker red band shows the 90% confidence intervals, whereas the lighter red band shows the 95% confidence intervals. The standard errors are clustered at the firm-pair level. Logged outcomes use $\log(1 + x)$. All specifications include calendar-year fixed effects and the baseline dyad-level controls as described in the text. The outcome is an indicator equal to one if, in year t , one firm cites the other at least once (extensive margin).

5.1.2 Origin and Destination

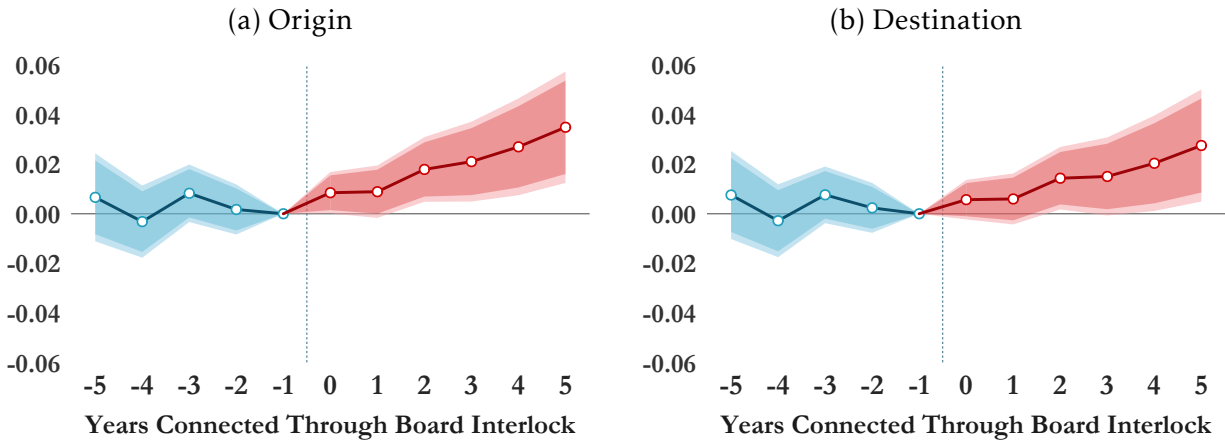
Next, I inspect whether these citation outcomes differ depending on the direction of director movement within the dyad. In particular, I define the origin firm as the firm in which the shared director was first appointed chronologically, and the destination firm as the company in the dyad that the director joins at a later date. For simplicity, I exclude dyads in which directors were appointed in the same year. This distinction allows me to ask whether the increase in cross-citation activity after the first interlock formation is concentrated on the side that exports the director or on the side that welcomes that director at a later time.

The results illustrate that the dynamic pattern is symmetric across origin and destination firms. The two panels of Figure 8 show that, after the first interlock, citations increase on both sides of the dyad. The effect only grows gradually over time, which again mirrors the same dynamic effects illustrated in Figures 6 and 7. The magnitude is very slightly larger on the origin side at longer horizons, but the two profiles are remarkably similar overall. Interlocks therefore seem to facilitate knowledge flows in both directions across the connected pair.

5.2 Innovation or Strategic Patenting?

It is difficult to judge from citation counts alone whether these results are consistent with technological learning or not. As a matter of fact, firms could use the same information obtained from newly-formed interlock links in two contrasting ways. They could either use this knowledge to start developing a more innovative patent portfolio, or use it to defend their existing products without producing new innovation. The former would be related to the IP leakage mechanism in [Cabezon and Hoberg \(2026\)](#), who show that dense overlapping-director networks among competitors accelerate the propagation of technologies across rivals. The latter would, for example, involve the use of strategic patenting ([Shapiro, 2000](#)). To understand which of these two mechanisms dominates, I first run a series of event studies that look at whether the patents and citations of newly interlocked dyads differ from the control group formed by never-interlocked dyads. Then, I analyze whether there are differences in how interlocked firms develop continuing patents.

Figure 8: Event Studies - Citations by Origin and Destination



Notes: Each panel reports staggered-adoption event study estimates of the effect of first adopting an interlock on the stated outcome. The reported estimates are from the [Callaway and Sant'Anna \(2021\)](#) framework, which maps cohort- and time-specific $ATT(g, t)$ to event time $k = t - g$, with $k = -1$ normalized to zero. Event time is defined relative to the first interlock ($k = 0$) between any two firms. The dots are the estimates from such regressions. The darker red band shows the 90% confidence intervals, whereas the lighter red band shows the 95% confidence intervals. The standard errors are clustered at the firm-pair level. Logged outcomes use $\log(1 + x)$. All specifications include calendar-year fixed effects and the baseline dyad-level controls as described in the text. The outcome is *Cross-Citations*, the share of a firm-year's patent citations that go to (or come from) firms with which the focal firm shares at least one director in year t , split between origin and destination firms as defined in the text.

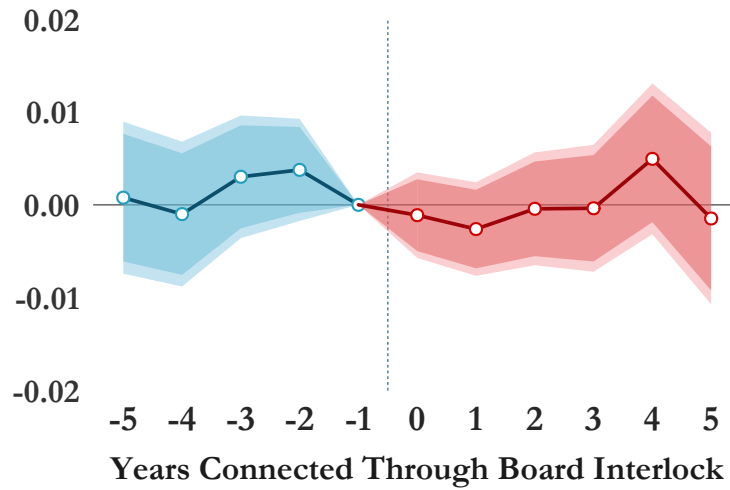
5.2.1 Event Studies

I start with new technological classes. For each unordered dyad $\{f, j\}$, I track the history of partner patents and partner technology classes cited by the pair up to year $t - 1$. I say that a citation in year t is classified as a new partner class citation if the cited patent belongs to at least one Cooperative Patent Classification (CPC) technology group that had not previously appeared among the dyad's cross-citations. One may be interested in analyzing these measures to understand whether interlocks facilitate exploration of previously unvisited parts of the partner's knowledge base. Figure 9 shows the results using this outcome measure. The results are flat: after two firms become interlocked for the first time, their patent citations do not start moving into new parts of the partner's technological space. The additional citation activity appears to remain within technology areas that were already familiar to the pair.

Next, I separate technologically proximate citations from non-technologically proximate ones. A citation is defined as technologically close when the citing patent and the cited patent share at least one technology group from the CPC classification system. This gives me a dyad-year share of citations to technologically close partner patents. I then compute the share of dyad-year citations directed to technologically and non-technologically close partner patents. If these cross-citation dynamics were indeed consistent with true innovation patterns, one would expect to see interlocked partners citing technologically close patents more often. In other words, the technological content of the citing patent should somewhat overlap with the technological domain of the cited partner patent if the innovation dynamics we see were indeed consistent with true technological learning. If the issuing patent were to cite partner patents that are completely outside its technological domain, the results would be harder to reconcile with the learning theory.

Figure 10 shows the results using these outcomes. In particular, Figure 10a tells us that, after first interlock formation, the two firms do not start citing each other more within overlapping technological areas. Instead, as Figure 10b points out, they start citing each other more outside their overlapping technology space. It is important to distinguish this exercise from the one reported in Figure 9, as the two sets of outcomes capture different margins. For instance, a citation can fall outside the dyad's overlapping technology space while still pointing to a partner technology class that the dyad had cited previously. Suppose that, before year t , the dyad had already cited partner patents in CPC class H01,

Figure 9: Event Studies - New Classes

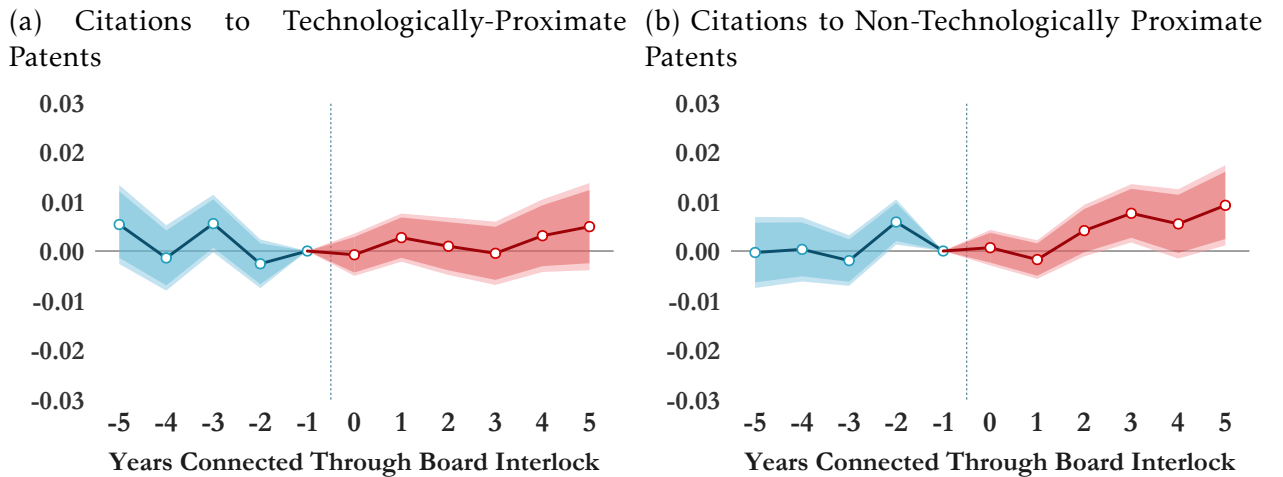


Notes: This figure reports staggered-adoption event study estimates of the effect of first adopting an interlock on the stated outcome. The reported estimates are from the [Callaway and Sant'Anna \(2021\)](#) framework, which maps cohort- and time-specific $ATT(g, t)$ to event time $k = t - g$, with $k = -1$ normalized to zero. Event time is defined relative to the first interlock ($k = 0$) between any two firms. The dots are the estimates from such regressions. The darker red band shows the 90% confidence intervals, whereas the lighter red band shows the 95% confidence intervals. The standard errors are clustered at the firm-pair level.

which covers electric elements. If, in year t , one firm cites an H01 partner patent from one of its own patents classified in G06, which covers computing and calculating technologies, the citation would not be technologically close, even though H01 is not a newly cited partner class. The patterns in Figures 9 and 10 are therefore consistent, as interlocks may increase citations across technologically distant areas without necessarily expanding the set of partner technology classes that firms cite.

I then look at the age of the cited patents. Do interlocked partners start citing older partner patents (defined as those issued at least 10 years prior) more frequently than newer patents? Figure 11 shows again that the answer is yes. Interlocks are associated with an increase in citations to older partner patents. Citations going to older patents is a trend that is difficult to square with the idea that interlocks enable firms to absorb the partner's frontier technological knowledge. It is instead more consistent with firms becoming more attentive to the partner's patent portfolio.

Figure 10: Event Studies - Technological Proximity

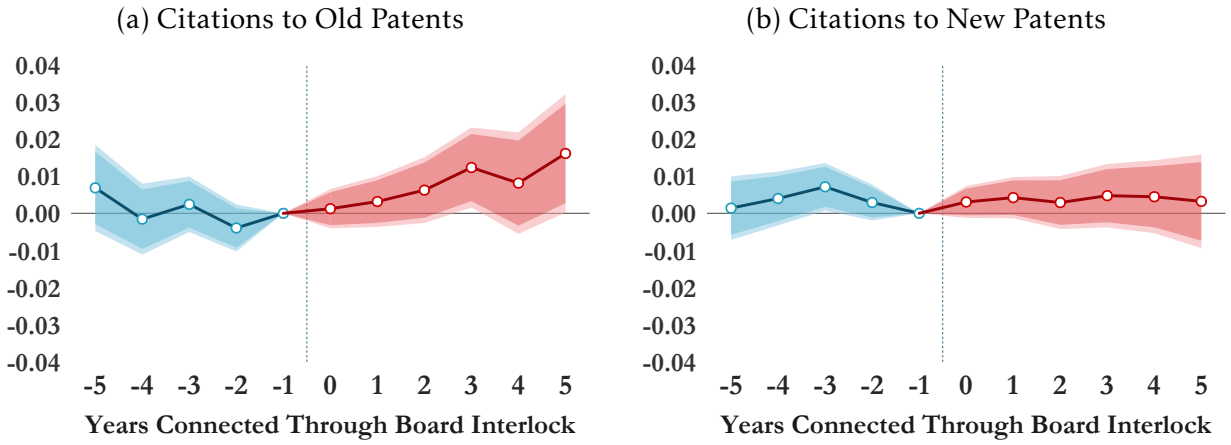


Notes: Each panel reports staggered-adoption event study estimates of the effect of first adopting an interlock on the stated outcome. The reported estimates are from the Callaway and Sant'Anna (2021) framework, which maps cohort- and time-specific $ATT(g, t)$ to event time $k = t - g$, with $k = -1$ normalized to zero. Event time is defined relative to the first interlock ($k = 0$) between any two firms. The dots are the estimates from such regressions. The darker red band shows the 90% confidence intervals, whereas the lighter red band shows the 95% confidence intervals. The standard errors are clustered at the firm-pair level. Logged outcomes use $\log(1+x)$. All specifications include calendar-year fixed effects and the baseline dyad-level controls as described in the text.

Finally, I analyze whether the cited patents are also more prominent. In particular,

I record whether, as of year $t - 1$, the cited patent has accumulated citations in the top decile of its patent-year-by-technology cohort. This component is constructed using only information available up to the year preceding the citation, so that the measure is predetermined at the time the citation occurs. Figure 12 shows that interlocks are associated with slightly higher citations towards these top decile patents, with the coefficients rising over time.

Figure 11: Event Studies - Patent Age



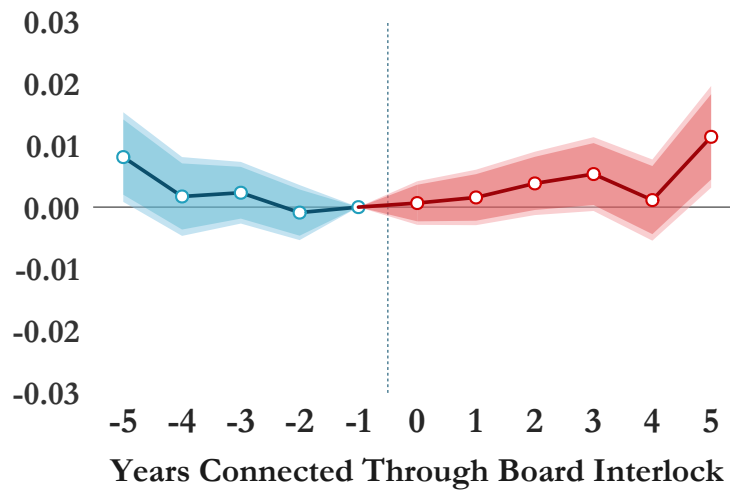
Notes: Each panel reports staggered-adoption event study estimates of the effect of first adopting an interlock on the stated outcome. I use the [Callaway and Sant'Anna \(2021\)](#) estimator, mapping cohort- and time-specific $ATT(g, t)$ to event time $k = t - g$, with $k = -1$ normalized to zero. The dots are the estimates from such regressions. The darker red band shows the 90% confidence intervals, whereas the lighter red band shows the 95% confidence intervals. The standard errors are clustered at the firm-pair level. Logged outcomes use $\log(1 + x)$. A patent is “old” if it is at least ten years old at the time of citation, and “new” otherwise. All specifications include calendar-year fixed effects and the baseline dyad-level controls described in the text.

To summarize, I find that after the formation of a new interlocking directorate, interlocked partners start citing each other’s patents more intensively. But these citations are disproportionately concentrated in favor of partner patents that are within already-familiar CPC technology classes, less technologically proximate, older, and in the top decile of most cited patents in their patent-year-by-technology cohort.

5.2.2 Continuing Patents

Inventors may file continuing applications, that is, patent child applications that extend the examination of the parent application. Continuations (CONs) pursue claims based on

Figure 12: Event Studies - Top Decile Patents



Notes: This figure reports staggered-adoption event study estimates of the effect of first adopting an interlock on the stated outcome. The reported estimates are from the [Callaway and Sant'Anna \(2021\)](#) framework, which maps cohort- and time-specific $ATT(g, t)$ to event time $k = t - g$, with $k = -1$ normalized to zero. Event time is defined relative to the first interlock ($k = 0$) between any two firms. The dots are the estimates from such regressions. The darker red band shows the 90% confidence intervals, whereas the lighter red band shows the 95% confidence intervals. The standard errors are clustered at the firm-pair level. Logged outcomes use $\log(1 + x)$. All specifications include calendar-year fixed effects and the baseline dyad-level controls as described in the text.

the same disclosure as the parent, continuations-in-part (CIPs) add new claims, and divisionals (DIVs) separate inventions that were jointly disclosed in the original application. By allowing applicants to postpone or revise claims, continuing applications help applicants deal with legal, technological, and commercial uncertainty. Nevertheless, this flexibility may also promote strategic patenting behavior and has been associated with greater litigation and overlapping patent rights that can impede competitor firms (Berger, Blind and Thumm, 2012, Hegde, Mowery and Graham, 2009, Righi and Simcoe, 2023, Righi, Cannito and Vladasel, 2023).

CIPs are the continuation type most closely associated with additional inventive activity, as patents in this category necessarily must contain additional technical disclosure relative to their parent application (see USPTO (2024) §§ 201.06–201.08). For instance, Hegde, Mowery and Graham (2009) show that CIPs are disproportionately used by R&D-intensive firms, particularly in chemical and biotechnological technologies. Of course, one should be careful in stating that CIPs represent innovations. A CIP establishes that the applicant has added new matter to the parent application, but that does not indicate that such new matter represents a major technological advance. Nevertheless, among the continuation types, CIPs offer the strongest signal of incremental innovation with respect to the parent application.

On the other hand, in a CON application, the applicant is legally prohibited from adding new matter (USPTO, 2024). In fact, the USPTO defines it as another application for inventions already disclosed in the parent. What can change through a continuation application is the set of claims, and therefore the legal scope of protection sought over that pre-existing disclosure. This is the reason why recent empirical work in Economics views CONs through the lens of strategic patenting behavior (Righi and Simcoe, 2023, Righi, Cannito and Vladasel, 2023). Recent work also documents that continuation applications are followed by a 37%-47% increase in the probability of litigation, with stronger effects when the continuation's claims differ substantially from those of its parents (Righi, 2026).

Finally, DIV applications are somewhat ambiguous for several reasons. To begin, DIVs cannot add new matter, and they only carve an independent or distinct invention out of subject matter that was already disclosed in the parent. But, unlike CONs, DIVs are often generated by the USPTO examination process itself. This happens when the examiner tells the applicant that the application contains multiple independent or distinct inven-

tions and that they must be prosecuted separately. DIVs cannot, therefore, be interpreted as evidence of active strategic patenting behavior. Nevertheless, [Hegde, Mowery and Graham \(2009\)](#) report that both CON and DIV applications tend to be associated with less important patents and interpret their use as consistent with strategic patenting behavior. Both because of this reason and because of transparency, I also report the results concerning DIVs alongside CONs and CIPs in the regressions below.¹²

Therefore, we want to look at whether the composition of these patents changes. For example, does the number of CONs, CIPs, and DIVs as a share of continuing patents change differently for interlocked firms? Table 9 illustrates the answers to this question. First, from column (1), we can see that the share of CONs is around 5.3 points higher for interlocked firms when purged of their firm, industry-year, and state-year effects, while column (3) says that the share of CIPS is lower by 4.5 points for interlocked firms according to the same specification. These results lose their statistical significance if, instead, we control for only the year and firm fixed effects. The fact that the pattern emerges only after absorbing industry-year and state-year effects points to the fact that aggregate differences across industries or locations mask the within-industry and within-state effect associated with interlocks. Finally, DIVs have a negative but statistically insignificant point estimate. According to the commonly held views about these types of continuing patents in the literature, these results suggest that interlocked firms pursue more continuing patents. However, those continuing patents contain, on average, less additional inventive activity and more applications consistent with strategic patenting compared to non-interlocked firms.

¹²Note that these categories are not mutually exclusive: while the vast majority of continuing applications are associated with only one type, some continuing applications may be linked to more than one continuing application type.

Table 9: Composition of Continuing Applications and Section 8 Interlock Exposure

	(1) CON / Continuing	(2) CON / Continuing	(3) CIP / Continuing	(4) CIP / Continuing	(5) DIV / Continuing	(6) DIV / Continuing
Interlock	0.053** (0.023)	0.032 (0.022)	-0.045* (0.025)	-0.005 (0.021)	-0.030 (0.023)	-0.035 (0.021)
N	1,854	2,133	1,854	2,133	1,852	2,130
R-squared	0.632	0.512	0.577	0.433	0.625	0.480
Firm FE	YES	YES	YES	YES	YES	YES
Year FE	NO	YES	NO	YES	NO	YES
Industry-Year FE	YES	NO	YES	NO	YES	NO
State-Year FE	YES	NO	YES	NO	YES	NO

Notes: *, **, and *** denote $p < 0.10$, $p < 0.05$, and $p < 0.01$, respectively. Standard errors (clustered by firm) are reported in parentheses. The unit of observation is the firm-year. *Interlock* is an indicator equal to one if the firm has at least one Section 8-violating horizontal interlock in year t and zero otherwise. All specifications include a set of firm-level controls chosen by double LASSO, the estimated firm-level AKM effects, and the corresponding sets of fixed effects as reported in the table.

6 Robustness Checks

I assess the robustness of my findings with a battery of robustness exercises reported in the Appendix section. In Section A.2, I report the sensitivity analyses from Oster (2019) and Diegert, Masten and Poirier (2025), which complement the Cinelli–Hazlett test reported in the main text. These additional analyses confirm that the results are robust to omitted variable bias. Then, in Appendix Section A.3, I report typical robustness checks to test the identifying assumption of my AKM model. In particular, I test the no-match effects assumption, as well as some mobility diagnostics. Both confirm that the identifying assumptions of the canonical AKM model are respected in my board member data.

In Appendix A.4, I assess the robustness of my event studies by reporting their placebo versions. In each event study, I assign each treated pair a fake treatment date and then re-estimate the same staggered adoption design. I find that these placebo event studies do not reproduce the post-treatment dynamic effects illustrated in the main text. Thus, this strengthens the interpretation that the observed increases in cross citations and extensive margin citation links are tied to actual interlock formation, and that the event study results are not generated by generic pre-existing trends around eventually treated firms.

7 Conclusion

This paper studies whether shared board members between competitors reshape firm-level innovation and, if so, through which channels. I find that firms engaged in this practice produce more patents, including citation-weighted patent counts, but do not display corresponding increases in R&D expenditure, R&D intensity, or market-based patent values. Therefore, interlocks do not primarily operate by inducing firms to spend more on formal research, but they appear to alter how firms convert existing innovative effort into observable patent output. The sensitivity analyses indicate that these results are very robust to residual omitted variable bias and that overturning the baseline patenting estimates would require unobservables to be implausibly strong relative to the rich set of observables already included in the specifications.

Event studies show that when two firms interlock for the first time, they begin citing each other more often in their patents. Nevertheless, the composition of those citations is not consistent with a story of technological learning or exploratory knowledge recombination: post-interlock cross citations are not disproportionately directed toward newer partner patents or technologically proximate inventions, as one would probably expect under technological learning. Instead, the stronger responses are concentrated in citations to partner patents that are older, more prominent – i.e., more cited in their technological cohort –, and not technologically close to the citing patent. In other words, interlocks do seem to transmit information across connected competitors, but the evidence presented in this paper suggests that this information is used primarily to sharpen firms' awareness of each other's patent boundaries and to support more strategic or defensive patenting behavior rather than to facilitate transmission of frontier technical knowledge.

The results concerning continuing patent applications reinforce this interpretation. On average, conditional on applying for a continuing patent, I find that interlocked firms shift the composition of these continuing patents more toward continuation patents – which are commonly associated with strategic patenting behavior –, while continuation-in-part patents – which usually reflect true innovation, as they add new technological specifications to the original claims – decrease as a share.

These results may carry some welfare and policy implications. If interlocks promoted knowledge diffusion and transmission of know-how between firms, then an increase in

patenting could be interpreted as at least partly innovation-enhancing. On the other hand, the results presented here provide evidence that the increase in post-interlock patent activity mainly reflects strategic patenting, which may for instance raise legal congestion in overlapping technology space. From an antitrust perspective, this may imply that the costs of interlocking directorates extend beyond the standard concerns about softened product-market competition or explicit coordination (Herrera-Caicedo, Jeffers and Prager, 2026): shared board members may also alter how firms defend incumbent positions, monitor rivals' intellectual-property portfolios, and shape the legal environment in which innovation happens in the first place. With that being said, to calculate the net welfare effects of this practice we would also need to take into account the effects on the consumers, which is beyond the scope of this paper.

The findings presented here reinforce the traditional view that interlocking directorates are organizational channels through which firms can reallocate effort and adjust strategic behavior (Allen, 1974, Bunting, 1976, Burt, 1983, Pfeffer, 1972, Pfeffer and Salancik, 1978, Stearns and Mizruchi, 1993) and offer a new mechanism through which this practice can alter innovation outcomes. This perspective also opens several avenues for future work. One of them, as I mentioned above, would be to undertake an assessment of the net welfare effect of this practice, which would need to take the effects on consumers into account in a structural model. Another would be to embed interlock formation in a structural network framework that permits counterfactual policy analysis of alternative enforcement regimes. Finally, future research could analyze whether interlocked firms experience lower litigation risks as a result of their new patenting behavior.

Understanding the effects of interlocking directorates requires looking not only at prices, output, and market structure, but also at how board connections reshape the legal and strategic margins of innovation. There is still a lot we do not understand about interlocking directorates, but the possibilities for future research seem ripe and promising.

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Appendix

A.1 Section 8 (Clayton Act) Classification of Horizontal Interlocks

This appendix describes, step by step, how I estimate Section 8 illegality for horizontal interlocking directorates. I aim to construct (i) a pair-year indicator for whether an observed competitor interlock is Section 8-violating and (ii) firm-year aggregates that count the number of such violations.

A.1.1 Scope and Legal Simplifications

The Section 8 screen from the Clayton Act is applied only to interlocks between firms that are classified as competitors in a given year, operationalized as belonging to the same industry. Vertical interlocks are not illegal by construction. Further, I implement the absolute de minimis exemption on competitive sales using the FTC’s annual dollar threshold. Section 8 contains exclusions that are specific to certain entities in some settings, and in practice financial institutions are treated differently. In my implementation, I therefore exclude banks and trust entities by setting $\text{Excluded}_f = 1$.

A.1.2 FTC Annual Thresholds

For each year t , I use the FTC’s annual Section 8 threshold table containing (i) the CSUP (capital, surplus, and undivided profits) coverage threshold, denoted $\overline{\text{CSUP}}_t$, and (ii) the absolute de minimis dollar threshold on competitive sales, denoted $\overline{S}_t$. Figure [A.1.1](#) plots these two threshold series over time (in millions of US dollars) for the years used in the empirical implementation. The CSUP threshold determines whether a firm is covered by Section 8 in a given year, while the competitive sales threshold is used to implement the absolute de minimis sales exemption.

A.1.3 Firm-Year Eligibility Inputs from Compustat

I construct a firm-year proxy for CSUP using Compustat data. Let $\widehat{\text{CSUP}}_{it}$ denote this proxy for firm i in year t . It is defined using a priority rule: (i) stockholders' equity if available, else (ii) common equity plus preferred stock when available, else (iii) total assets minus total liabilities when available. Then, let S_{it} denote firm-year net sales. I define the two threshold indicators:

$$\text{Covered}_{it} = \mathbb{1}\{\widehat{\text{CSUP}}_{it} \geq \overline{\text{CSUP}}_t\}, \quad \text{ExemptAbs}_{it} = \mathbb{1}\{S_{it} < \bar{S}_t\}.$$

A firm-year is eligible for the interlocking directorate restrictions according to Section 8 of the Clayton Act if it is covered and not exempt under the absolute sales cutoff, that is,

$$\text{Eligible}_{it} = \mathbb{1}\{\text{Covered}_{it} = 1 \wedge \text{ExemptAbs}_{it} = 0\}.$$

This is defined only for the firms' observations for which the underlying accounting inputs and thresholds are observed.

A.1.4 Pair-Year Legality for Horizontal Interlocks

Let $\text{Interlock}_{fjt} \in \{0, 1\}$ indicate whether firms f and j share at least one board member in year t . Consider a horizontal competitor pair-year. The interlock is classified as Section 8-violating if an interlock exists, both firms are eligible, and neither firm is excluded:

$$\text{Sec8Violate}_{fjt} = \mathbb{1}\{\text{Interlock}_{fjt} = 1\} \cdot \mathbb{1}\{\text{Eligible}_{ft} = 1\} \cdot \mathbb{1}\{\text{Eligible}_{jt} = 1\} \quad (\text{A.1.1})$$

$$\cdot \mathbb{1}\{\text{Excluded}_f = 0\} \cdot \mathbb{1}\{\text{Excluded}_j = 0\}, \quad (\text{A.1.2})$$

where Sec8Violate_{fjt} is only defined on the restricted sample of horizontal pairs.

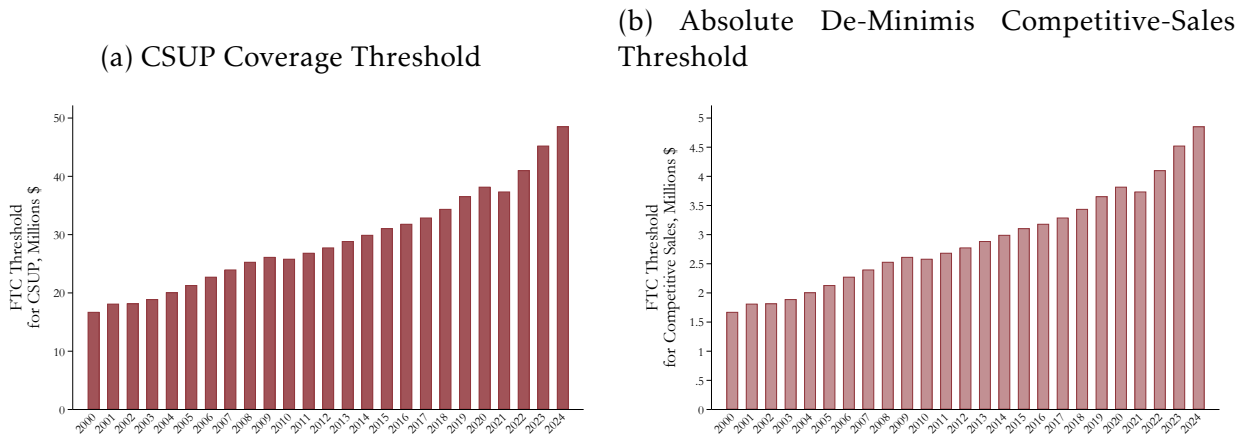
A.1.5 Aggregation to Firm-Year Violation Counts

To obtain firm-year counts, I first sum pair-year violations across all competitor pairs before collapsing to a unique firm-year:

$$\text{Sec8Violate}_{jt} = \sum_{j \neq f} \text{Sec8Violate}_{fjt}.$$

This gives me a firm-year panel of Section 8–violating horizontal interlocks. Then, in my baseline regressions, I use an indicator variable Interlock_{jt} that is equal to one if firm j has at least one horizontal interlock at time t , that is, $\text{Sec8Violate}_{jt} \geq 1$.

Figure A.1.1: FTC Annual Section 8 Coverage Thresholds Used in the Section 8 Screen



Notes: Panel (a) plots the FTC’s annual threshold for “capital, surplus, and undivided profits” (CSUP), and Panel (b) plots the FTC’s annual absolute de-minimis “competitive sales” threshold. Both are shown in millions of dollars for the years used in the implementation window (2000–2024). These thresholds are used to (i) determine firm-year Section 8 coverage via CSUP and (ii) implement the absolute sales exemption. They enter the pair-year legality screen described in Appendix A.1.

A.2 Sensitivity Analyses

The modern literature on sensitivity analyses in Economics begins, to my knowledge, with [Altonji, Elder and Taber \(2005\)](#). Studying the effect of Catholic high school attendance on subsequent college outcomes, they develop a framework for assessing the importance of omitted variable bias in settings where identification does not rest on a clear exogenous shock or a valid instrument. Their idea is to compare selection on unobservables with selection on observables after conditioning on a rich set of covariates.

To formalize this idea, let $\hat{\beta}^S$ denote the coefficient on the interlock variable from a short regression of Y_{it} on the treatment only, and let $\hat{\beta}^M$ denote the coefficient from a medium regression that adds the observed controls W_{it} . Let R_S^2 and R_M^2 be the corresponding R^2 values. In [Altonji, Elder and Taber \(2005\)](#), the relevant statistic is the ratio

$$\text{AET} = \frac{\hat{\beta}^M}{\hat{\beta}^S - \hat{\beta}^M},$$

which compares the coefficient that remains after controlling for observables to the amount by which the coefficient shrinks once observables are added. A larger value means that selection on unobservables would have to be large, relative to selection on observables, to drive the coefficient of interest to zero. While this approach is useful and intuitive, it does not scale coefficient movements by the explanatory power of the added controls. [Oster \(2019\)](#) builds on and sharpens this idea by combining the movement in the treatment coefficient with the movement in R^2 . Under proportional selection, and given a hypothesized upper bound $R_{\max}$ for the R^2 of the infeasible long regression, Oster's bias-adjusted coefficient is

$$\beta^*(\delta, R_{\max}) = \hat{\beta}^M - \delta(\hat{\beta}^S - \hat{\beta}^M) \frac{R_{\max} - R_M^2}{R_M^2 - R_S^2},$$

where δ measures the strength of selection on unobservables relative to selection on observables. Setting $\delta = 1$ imposes the benchmark assumption that selection on unobservables is as strong as selection on observables. Solving $\beta^*(\delta, R_{\max}) = 0$ yields the Oster breakdown value

$$\delta_0 = \frac{\hat{\beta}^M(R_M^2 - R_S^2)}{(\hat{\beta}^S - \hat{\beta}^M)(R_{\max} - R_M^2)},$$

that is, the value of δ required to drive the adjusted coefficient to zero.

More recently, [Diegert, Masten and Poirier \(2025\)](#) (DMP) develop a sensitivity analysis method that does not require the included controls to be exogenous. Let β^{long} denote the coefficient on the treatment in the long regression that includes both observed controls W_1 and omitted controls W_2 , and let β^{med} be the coefficient from the regression that includes only W_1 . Their paper characterizes an identified set for β^{long} as a function of sensitivity parameters that bound how strongly the omitted variables can predict the treatment and the outcome.

In the one-parameter version, let $\bar{r}_X$ bound the strength of selection on unobservables in the treatment equation relative to observables. Specifically, $\bar{r}_X$ bounds the ratio between the standard-deviation contribution of the unobserved covariate index and the corresponding contribution of the observed covariate index in the linear projection of X . Under this restriction, and provided that

$$\bar{r}_X < \sqrt{1 - R_{X \sim W_1}^2},$$

the identified set is

$$B(\bar{r}_X) = \left[\beta^{\text{med}} - \sqrt{\frac{\text{Var}(Y_{X, W_1}^\perp)}{\text{Var}(X_{W_1}^\perp)}} \sqrt{\frac{\bar{r}_X^2 R_{X \sim W_1}^2}{1 - R_{X \sim W_1}^2 - \bar{r}_X^2}}, \beta^{\text{med}} + \sqrt{\frac{\text{Var}(Y_{X, W_1}^\perp)}{\text{Var}(X_{W_1}^\perp)}} \sqrt{\frac{\bar{r}_X^2 R_{X \sim W_1}^2}{1 - R_{X \sim W_1}^2 - \bar{r}_X^2}} \right],$$

where $X_{W_1}^\perp$ illustrates the residual from projecting the treatment X on the observed controls W_1 , whereas $Y_{X, W_1}^\perp$ is the residual from projecting the outcome Y on X and W_1 . If $\bar{r}_X \geq \sqrt{1 - R_{X \sim W_1}^2}$, the corresponding bounds are unbounded. The term $R_{X \sim W_1}^2$ is the R^2 from the treatment equation that projects X on W_1 . I use the notation $R_{A \sim B \bullet C}^2$ to denote the partial R^2 from a regression of A on B , after partialling out C . $R_{Y \sim X \bullet W_1}^2$ is therefore the share of residual variation in Y (after removing W_1) that is explained by the residualized treatment $X_{W_1}^\perp$. The symbol $\bullet$ separates the variable whose additional explanatory power is being measured from the variables being held fixed.

This expression delivers a formula for sign robustness, too, because if the entire interval $B(\bar{r}_X)$ remains positive, then omitted variables whose treatment selection strength is at most $\bar{r}_X$ times the corresponding strength of the observed controls cannot flip the sign of the estimate. The associated sign-change breakdown point, corresponding to Corollary

1 in Diegert, Masten and Poirier (2025), is

$$\bar{r}_X^{bp} = \left(\frac{R_{Y \sim X \bullet W_1}^2}{\frac{R_{X \sim W_1}^2}{1 - R_{X \sim W_1}^2} + R_{Y \sim X \bullet W_1}^2} \right)^{1/2}.$$

This is the largest value of $\bar{r}_X$ for which the sign of the coefficient remains pinned down. Equivalently, if $\bar{r}_X^{bp} = k$, then omitted variables would have to be more than k times as strong as the observed controls, in terms of treatment selection, to overturn the sign of the estimate. DMP also delivers estimates for sign robustness when additional restrictions are imposed on how strongly omitted variables may affect the outcome equation. In the regression tables below, I report the Oster bias-adjusted coefficients and breakdown values, together with the DMP sign robustness diagnostics.

Another approach to assess sensitivity to omitted variable bias is provided by Cinelli and Hazlett (2020). They depart from the proportional selection logic underlying the approaches discussed above and instead parameterizes the strength of unobserved confounding directly through partial R^2 measures. To formalize their idea, consider the population regression

$$Y = \tau D + X'\beta + \gamma U + \varepsilon,$$

where D denotes the treatment or regressor of interest, X is the vector of observed controls, and U is an omitted confounder. Cinelli and Hazlett (2020) show that the magnitude of omitted-variable bias can be characterized in terms of two quantities:

$$R_{D \sim U|X}^2 \quad \text{and} \quad R_{Y \sim U|D,X}^2.$$

The first term measures the proportion of the residual variation in D , after conditioning on X , explained by the omitted confounder U . The second one denotes the proportion of the residual variation in Y , after conditioning on both D and X , explained by U . These two partial R^2 statistics therefore capture the treatment side and outcome side strength of the omitted confounding, respectively. The omitted variable bias in the estimated treatment coefficient can be written as a direct function of these two quantities: let $\widehat{\tau}_{\text{res}}$ denote

the coefficient obtained from the regression omitting U , and let $\widehat{\tau}$ denote the coefficient that would be obtained if U were observed and included. Then, the absolute bias satisfies

$$|\widehat{\tau}_{\text{res}} - \widehat{\tau}| = \text{se}(\widehat{\tau}_{\text{res}}) \sqrt{\text{df} \frac{R_{Y \sim U|D,X}^2 R_{D \sim U|X}^2}{1 - R_{D \sim U|X}^2}}.$$

Thus, any hypothesized pair

$$(R_{D \sim U|X}^2, R_{Y \sim U|D,X}^2)$$

implies a corresponding degree of omitted variable bias and, consequently, a bias-adjusted estimate of the treatment effect.

For a given proportional reduction q in the estimated coefficient, the robustness value from [Cinelli and Hazlett \(2020\)](#) is the minimum common partial R^2 that an omitted confounder would need to have with both the treatment and the outcome to generate such a reduction. In the commonly considered case with $q = 1$, the robustness value measures the minimum strength required from confounding variables to reduce the estimated treatment effect to zero. Analogously, RV_α measures the minimum confounding strength required to make the estimate statistically insignificant at a specified significance level. It is important to highlight that the restriction that the two partial R^2 values are equal is used only to construct this scalar summary. In practice, the underlying sensitivity analysis allows the treatment side and outcome side strength of confounding to differ.

Relative to [Oster \(2019\)](#) and [Diegert, Masten and Poirier \(2025\)](#), [Cinelli and Hazlett \(2020\)](#) characterize confounding directly through the explanatory power that an omitted variable would need to have for the residualized treatment and residualized outcome. This allows the econometrician to express their sensitivity parameters to be benchmarked directly against the partial R^2 values associated with observed covariates, which allows for easy interpretation. One can for example ask whether an omitted confounder would need to be as strongly associated with treatment and outcome as a particular observed covariate, or several times stronger, to overturn the estimated effect.

The framework is nevertheless closely connected to the approaches discussed above. For example, [Oster \(2019\)](#)'s R_{max} can be mapped into the outcome side of the [Cinelli and](#)

Hazlett (2020) sensitivity parameter according to

$$R_{Y \sim U|D,X}^2 = \frac{R_{\max} - R_{Y \sim D,X}^2}{1 - R_{Y \sim D,X}^2}.$$

The principal conceptual difference lies in the treatment side. While proportional selection approaches impose structure linking selection on unobservables to selection on observables, Cinelli and Hazlett (2020) leave

$$R_{D \sim U|X}^2$$

as a sensitivity parameter to be calibrated directly. Hence, their method does not require the researcher to posit a particular proportional relationship between observed and unobserved selection, like the Oster (2019) test does.

Further, Diegert, Masten and Poirier (2025) provide a framework in which the strength of unobserved selection can still be expressed relative to observed selection while explicitly allowing such correlation and deriving corresponding identified sets. On the other hand, Cinelli and Hazlett (2020) directly characterize how strong the residual association between the omitted confounder, treatment, and outcome must be in order to alter the estimated coefficient.

The Cinelli and Hazlett (2020) analysis should therefore be viewed as complementary to the Oster (2019) and Diegert, Masten and Poirier (2025) exercises, and reporting all these types of sensitivity analysis provides robustness assessments under distinct, but closely related, ways of formalizing the potential importance of unobserved confounding. Accordingly, in the main part of the paper, I report the sensitivity analysis test from Cinelli and Hazlett (2020). In this Appendix, Table A.2.1 reports the sensitivity results for innovation outputs using the Oster (2019) and Diegert, Masten and Poirier (2025) tests. It indicates that these estimates are unlikely to be driven by omitted variable bias. The corresponding Oster β values are 0.019 and 0.038 for the raw patent counts, implying that the coefficients remain positive even under the benchmark assumption that selection on unobservables is as strong as selection on observables. The Oster- δ values, 1.311 and 1.491, imply that selection on unobservables would need to be almost 1.5 times as strong

as selection on observables to drive the corresponding coefficients to zero.¹³ The DMP breakdown values, 105.706 and 52.211, point in the same direction: omitted variables would need to be remarkably strong, relative to the observed controls, before the sign of the estimates would no longer be guaranteed. The Oster- δ numbers for citation-weighted patent counts also sit close to one, pointing to the fact that unobservables would need to be almost as strong as the observed covariates to drive these estimates to zero. The corresponding DMP figures are robust.

Table A.2.1: Sensitivity Analyses on the Effects of Horizontal Interlocks on Innovation Outputs

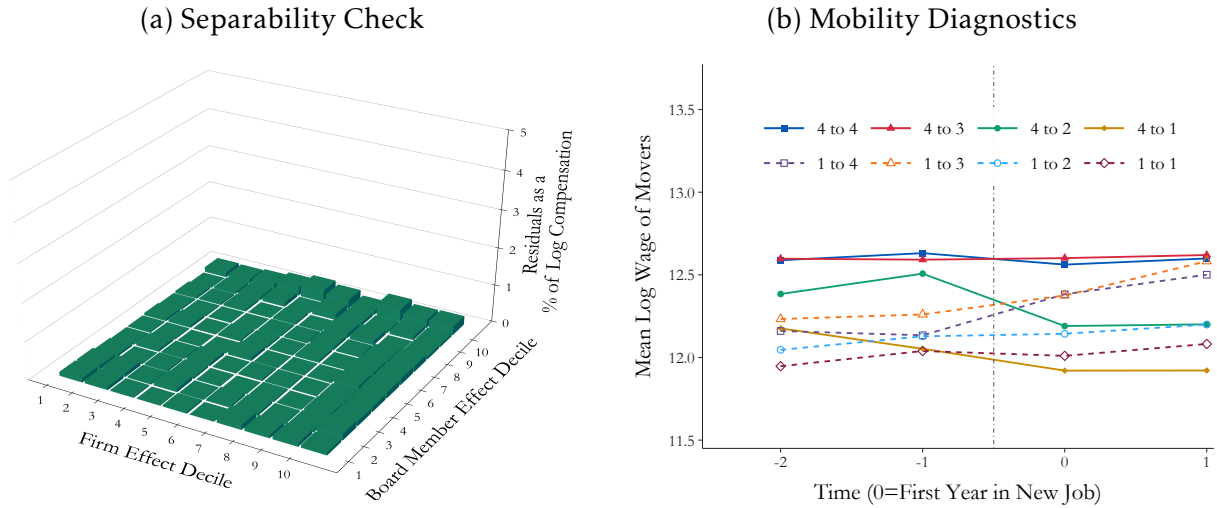
	(1) Log(Patents)	(2) Log(Patents)	(3) Log(CW Patents)	(4) Log(CW Patents)
Oster- β	[0.019]	[0.038]	[-0.006]	[-0.013]
Oster- δ	{1.311}	{1.491}	{0.924}	{0.872}
DMP Breakdown	<105.706>	<52.211>	<11.301>	<11.516>
N	3,889	4,121	3,933	4,165
R-squared	0.936	0.920	0.898	0.871
Firm FE	YES	YES	YES	YES
Year FE	NO	YES	NO	YES
Industry-Year FE	YES	NO	YES	NO
State-Year FE	YES	NO	YES	NO

Notes: Each column reports the sensitivity of the *Interlock* coefficient of the corresponding OLS specification to unobserved confounding. Oster- β is the Oster (2019) bias-adjusted coefficient at $\delta = 1$ and a maximum R^2 of 1.3 times the observed one; Oster- δ is the value of δ that would set that coefficient to zero at the same maximum R^2 ; and DMP Breakdown is the [Diegert, Masten and Poirier \(2025\)](#) breakdown point beyond which the sign of the effect can no longer be signed.

¹³For reference, [Oster \(2019\)](#) reports that only 45% of the non-randomized papers from the top 5 journals in Economics have an Oster delta greater than 1.

A.3 AKM Model Diagnostics

Figure A.3.1: Diagnostics of the AKM Model



Notes: Panel A splits the estimated AKM board member and firm fixed effects into deciles and then, for each of the one hundred cells, reports their mean residuals (in % of log compensation). Panel B presents an event-study mobility test that categorizes companies by quartiles of their mean log wages and then classifies each job change by the quartile of the origin and destination workplaces. Both tests are from [Card, Heining and Kline \(2013\)](#).

In this Appendix section, I report common tests that have been developed to test whether the assumptions of the baseline AKM model are respected. To begin, Figure A.3.1 reports graphical tests from [Card, Heining and Kline \(2013\)](#). In particular, Figure A.3.1a evaluates the additively separable assumption of the canonical AKM model to exclude match effects between the workers and the firms. In this test, both the board member and the firm fixed effects are split into deciles. Then, for each of the one hundred different cells, I compute the average size of the residuals as a percentage of the logarithm of the compensation for the board members in that cell. A flat distribution would be evidence against economically important match effects and is consistent with the additively separable structure of director and firm fixed effects. The results from Figure A.3.1a show that, in my estimated board member model, the distribution of these residuals is extremely flat and does not show evidence of systematic patterns across deciles.

Figure A.3.1b presents a mobility diagnostic for the same model. The figure traces

mean log compensation in event time for directors who move between boards, grouping moves by the quartile of the mean coworker compensation at the origin and destination boards. The exogenous mobility assumption implies that, absent important match-specific or transitory shocks driving moves across firms, compensation profiles should be relatively stable before the transition and should exhibit a discrete change at the move whose sign and magnitude vary systematically with the direction of mobility. This is what Figure [A.3.1b](#) shows. Directors who move from low-pay to higher-pay companies experience upward shifts in compensation at the time of the move, with larger gains for moves to higher destination quartiles, whereas directors who move from high-pay boards to lower-pay boards display corresponding declines. Moves within the same quartile generate modest changes. These visual results are consistent with the interpretation that the estimated board component captures persistent differences in pay levels across boards.

A.4 Robustness Checks

A.4.1 Placebo Event Studies

To assess whether the patterns in the staggered event studies illustrated in the main paper could be generated mechanically by generic trends around future interlocks rather than by interlocks themselves, I estimate placebo versions of these event studies. If the baseline results are truly tied to the timing of interlock formation, then assigning treatment to an earlier fake date should not reproduce the same dynamics.

For the dyad-level citation outcomes, I begin from the same firm-pair-year panel used in the main event studies and construct pseudo-treatment dates by shifting each treated dyad’s true first-interlock year backward by six years. This meant that, if G_{ij} denotes the first year in which dyad (i, j) becomes interlocked, I define the placebo cohort as

$$G_{ij}^P = G_{ij} - 6$$

for treated dyads, while never-treated dyads remain untreated. This is the same placebo event study framework discussed in [Borusyak, Jaravel and Spiess \(2024\)](#) and a common way to generate placebo event studies. Dyads whose placebo treatment year would fall before the beginning of the usable event study window are excluded. I then re-estimate the same staggered-adoption [Callaway and Sant’Anna \(2021\)](#) design as in the main text, replacing the true treatment date with G_{ij}^P and keeping the same event-time structure, with $k = -1$ as the omitted period.

For each placebo outcome $Y_{ij,t}$, I first estimate group-time average treatment effects,

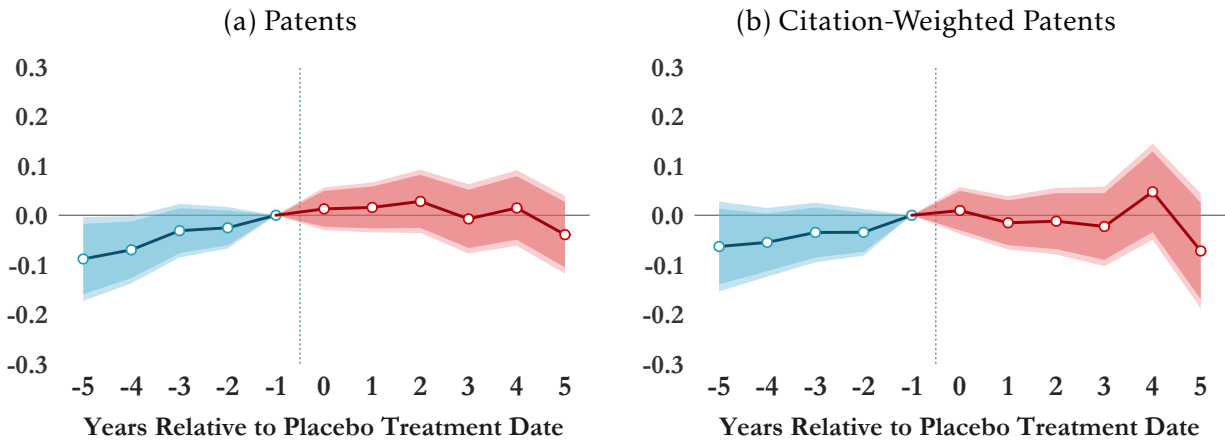
$$ATT^P(g, t) = \mathbb{E}[Y_{ij,t}(g) - Y_{ij,t}(\infty) \mid G_{ij}^P = g],$$

where g indexes the placebo treatment cohort and $Y_{ij,t}(\infty)$ denotes the untreated potential outcome. These group-time effects compare dyads assigned to placebo cohort g with dyads that are untreated at t . I then aggregate the cohort-time effects by event time $k = t - g$:

$$\theta_k^P = \sum_{g: g+k \leq T} w_{g,k}^P ATT^P(g, g+k), \quad k \neq -1,$$

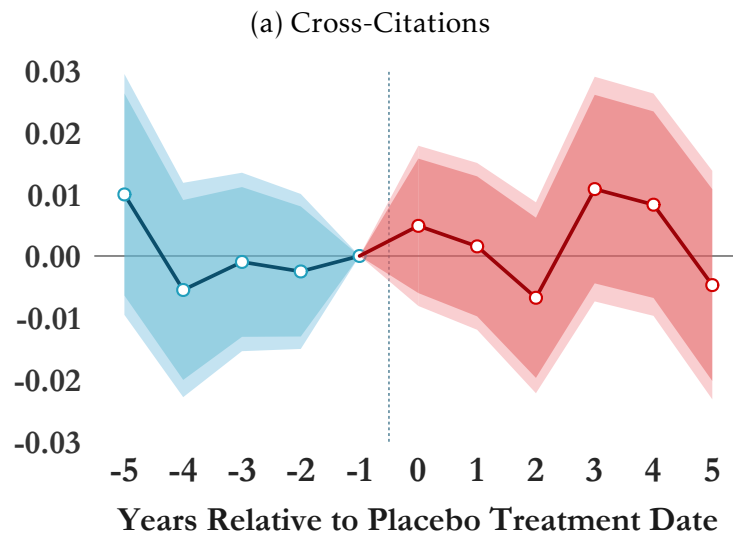
where $w_{g,k}^P$ are the [Callaway and Sant'Anna \(2021\)](#) aggregation weights, and the period $k = -1$ is normalized to zero. I apply this placebo procedure to patent counts, citations (including cross citations, the split by origin and destination, and the types of patents being cited), and the extensive margin cited dummy. Under the identifying interpretation of the main results, these placebo estimates should remain close to zero throughout the event window. Indeed, I find that they do, which strengthens the view that the main event study effects are linked to actual interlock formation rather than to slow-moving differences between eventually treated and untreated dyads.

Figure A.4.1: Placebo Event Studies - Patent Counts



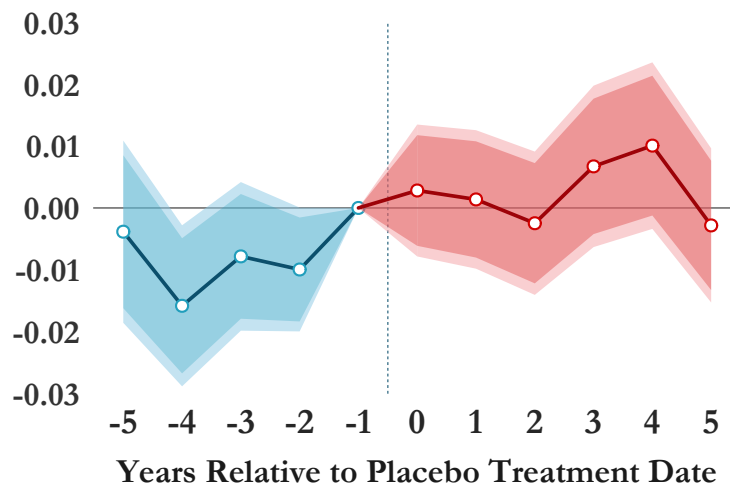
Notes: This figure reports staggered-adoption event study estimates of the effect of the placebo adoption date on the stated outcome. The reported estimates are from the [Callaway and Sant'Anna \(2021\)](#) estimator, which maps cohort- and time-specific $ATT(g, t)$ to event time $k = t - g$, with $k = -1$ normalized to zero. The dots are the estimates from such regressions. The darker red band shows the 90% confidence intervals, whereas the lighter red band shows the 95% confidence intervals. The standard errors are clustered at the firm-pair level. Event time is defined relative to the placebo treatment date ($k = 0$); $k = -1$ is the omitted baseline.

Figure A.4.2: Placebo Event Studies - Citations



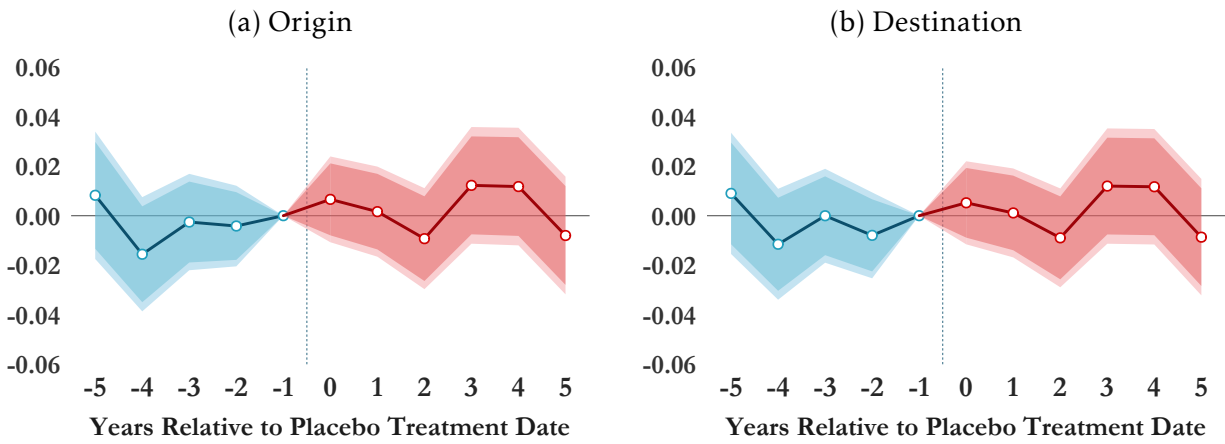
Notes: This figure reports staggered-adoption event study estimates of the effect of the placebo adoption date on the stated outcome. The reported estimates are from the [Callaway and Sant'Anna \(2021\)](#) estimator, which maps cohort- and time-specific $ATT(g, t)$ to event time $k = t - g$, with $k = -1$ normalized to zero. The dots are the estimates from such regressions. The darker red band shows the 90% confidence intervals, whereas the lighter red band shows the 95% confidence intervals. The standard errors are clustered at the firm-pair level. Event time is defined relative to the placebo treatment date ($k = 0$); $k = -1$ is the omitted baseline.

Figure A.4.3: Placebo Event Studies – Cited Dummy (Extensive Margin)



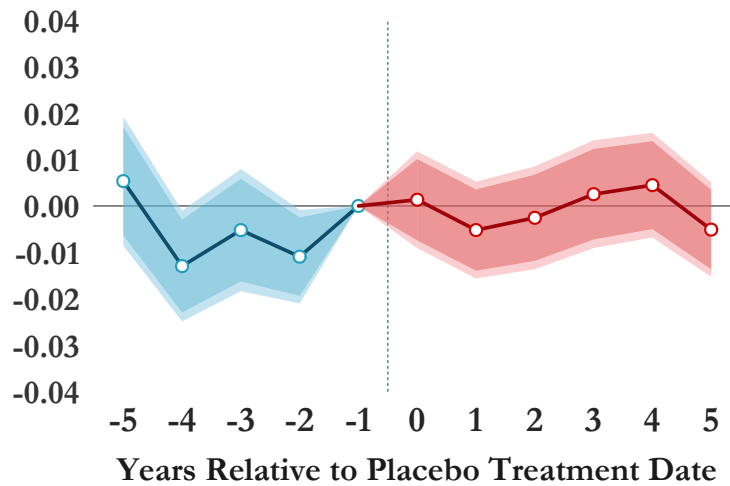
Notes: This figure reports staggered-adoption event study estimates of the effect of the placebo adoption date on the stated outcome. The reported estimates are from the [Callaway and Sant'Anna \(2021\)](#) estimator, which maps cohort- and time-specific $ATT(g, t)$ to event time $k = t - g$, with $k = -1$ normalized to zero. The dots are the estimates from such regressions. The darker red band shows the 90% confidence intervals, whereas the lighter red band shows the 95% confidence intervals. The standard errors are clustered at the firm-pair level. Event time is defined relative to the placebo treatment date ($k = 0$); $k = -1$ is the omitted baseline.

Figure A.4.4: Placebo Event Studies - Citations by Origin and Destination



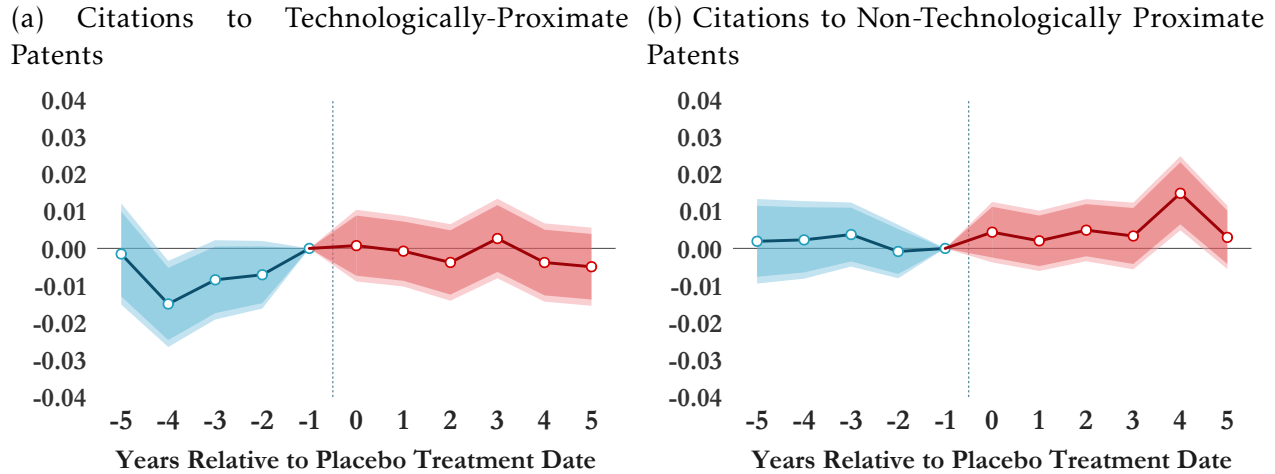
Notes: This figure reports staggered-adoption event study estimates of the effect of the placebo adoption date on the stated outcome. The reported estimates are from the [Callaway and Sant'Anna \(2021\)](#) estimator, which maps cohort- and time-specific $ATT(g, t)$ to event time $k = t - g$, with $k = -1$ normalized to zero. The dots are the estimates from such regressions. The darker red band shows the 90% confidence intervals, whereas the lighter red band shows the 95% confidence intervals. The standard errors are clustered at the firm-pair level. Event time is defined relative to the placebo treatment date ($k = 0$); $k = -1$ is the omitted baseline.

Figure A.4.5: Placebo Event Studies – New Classes



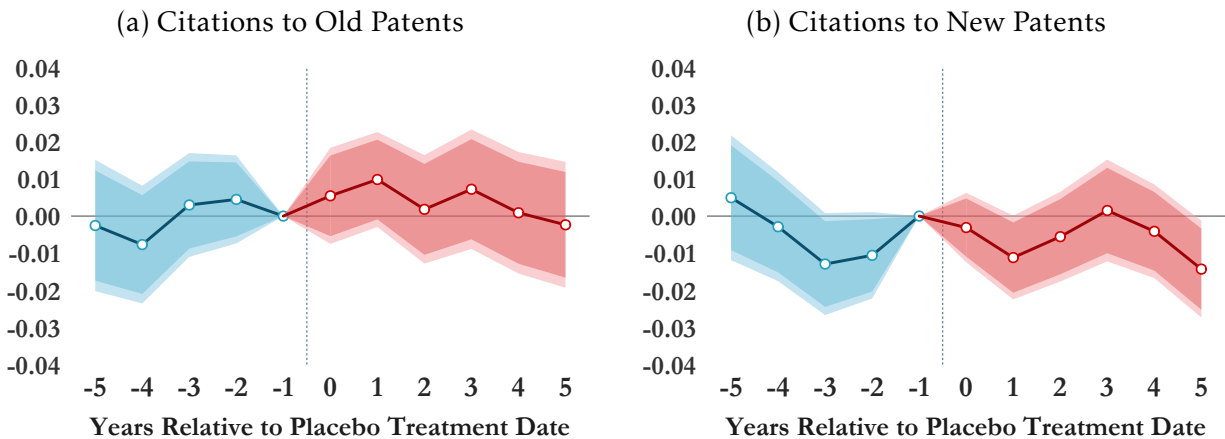
Notes: This figure reports staggered-adoption event study estimates of the effect of the placebo adoption date on the stated outcome. The reported estimates are from the [Callaway and Sant'Anna \(2021\)](#) estimator, which maps cohort- and time-specific $ATT(g, t)$ to event time $k = t - g$, with $k = -1$ normalized to zero. The dots are the estimates from such regressions. The darker red band shows the 90% confidence intervals, whereas the lighter red band shows the 95% confidence intervals. The standard errors are clustered at the firm-pair level. Event time is defined relative to the placebo treatment date ($k = 0$); $k = -1$ is the omitted baseline.

Figure A.4.6: Placebo Event Studies - Technological Proximity



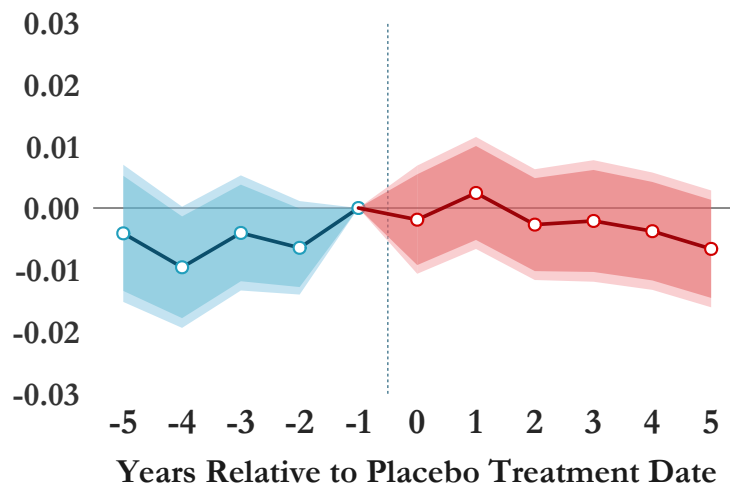
Notes: This figure reports staggered-adoption event study estimates of the effect of the placebo adoption date on the stated outcome. The reported estimates are from the [Callaway and Sant'Anna \(2021\)](#) estimator, which maps cohort- and time-specific $ATT(g, t)$ to event time $k = t - g$, with $k = -1$ normalized to zero. The dots are the estimates from such regressions. The darker red band shows the 90% confidence intervals, whereas the lighter red band shows the 95% confidence intervals. The standard errors are clustered at the firm-pair level. Event time is defined relative to the placebo treatment date ($k = 0$); $k = -1$ is the omitted baseline.

Figure A.4.7: Placebo Event Studies - Patent Age



Notes: This figure reports staggered-adoption event study estimates of the effect of the placebo adoption date on the stated outcome. The reported estimates are from the [Callaway and Sant'Anna \(2021\)](#) estimator, which maps cohort- and time-specific $ATT(g, t)$ to event time $k = t - g$, with $k = -1$ normalized to zero. The dots are the estimates from such regressions. The darker red band shows the 90% confidence intervals, whereas the lighter red band shows the 95% confidence intervals. The standard errors are clustered at the firm-pair level. Event time is defined relative to the placebo treatment date ($k = 0$); $k = -1$ is the omitted baseline.

Figure A.4.8: Placebo Event Studies - Top Decile Patents

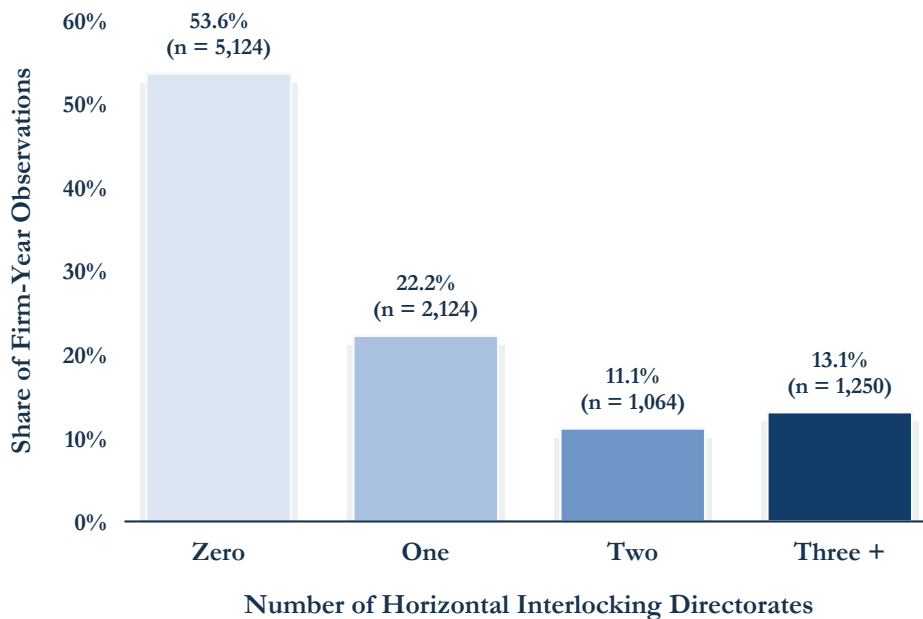


Notes: This figure reports staggered-adoption event study estimates of the effect of the placebo adoption date on the stated outcome. The reported estimates are from the [Callaway and Sant'Anna \(2021\)](#) estimator, which maps cohort- and time-specific $ATT(g, t)$ to event time $k = t - g$, with $k = -1$ normalized to zero. The dots are the estimates from such regressions. The darker red band shows the 90% confidence intervals, whereas the lighter red band shows the 95% confidence intervals. The standard errors are clustered at the firm-pair level. Event time is defined relative to the placebo treatment date ($k = 0$); $k = -1$ is the omitted baseline.

A.5 Number of Interlocking Directorates

In this section, I analyze whether the results change based on the total number of horizontal interlocking directorates that a firm j has. Figure A.5.1 begins by plotting the share of firm-year observations based on the number of horizontal interlocks. About half of firm-year observations have no interlocks. Then, conditional on interlocking, most firms only have one horizontal interlock (22.2% of total firm-year observations). In exactly 11.1% of firm-year observations in the sample we observe two interlocks. In 13.1% of cases we see three or more interlocks at the same time.

Figure A.5.1: Number of Horizontal Interlocking Directorates by Bracket



Notes: This figure reports the distribution of the number of Section 8—violating horizontal interlocking directorates across firm–year observations in the estimation sample. The four brackets correspond to zero, one, two, and three or more horizontal interlocks in year t . Percentages sum to 100, and counts in parentheses report the number of firm–year observations in each bracket.

Tables A.5.1–A.5.3 report the results for innovation inputs, outputs, and values by the number of interlocking directorates a firm has, where firms with zero interlocks are taken as the base group. Interestingly, the null effect on R&D expenditure reported in the main paper now displays a negative coefficient for firms with exactly two interlocks

and a positive one for firms with three or more interlocks, which offset each other. In the specification with only firm and year fixed effects, only the positive coefficient for firms with three or more interlocks is confirmed. R&D intensity remains null, except for a slightly negative coefficient for firms with exactly two interlocks in the specification with firm, industry-year, and state-year fixed effects. Finally, firms with two interlocks display a negative coefficient on the capital–labor ratio.

When it comes to innovation outputs, Table A.5.2 shows that the increase in patents is roughly homogeneous across the board. Finally, Table A.5.3 shows that there is a mechanical increase in the total value of patents across firms, which is expected given the increase in patent counts, although the null estimates on average value confirm that the average patent granted to interlocked firms does not carry higher market-based values.

Table A.5.1: Effects of Horizontal Interlock Brackets on Innovation Inputs and Production Structure

	(1) Log(R&D)	(2) Log(R&D)	(3) Log(R&D Int.)	(4) Log(R&D Int.)	(5) Log(Cap/Lab)	(6) Log(Cap/Lab)
1 Interlock	-0.009 (0.021)	-0.017 (0.022)	0.004 (0.003)	0.005 (0.003)	-0.035 (0.024)	-0.041 (0.027)
2 Interlocks	-0.108* (0.063)	-0.076 (0.064)	-0.018* (0.010)	-0.016 (0.010)	-0.103* (0.054)	-0.131*** (0.050)
3+ Interlocks	0.170* (0.091)	0.204* (0.105)	-0.004 (0.028)	-0.001 (0.028)	0.049 (0.094)	0.099 (0.070)
N	4,210	4,451	4,257	4,497	3,354	3,586
R-squared	0.990	0.987	0.861	0.846	0.972	0.964
Firm FE	YES	YES	YES	YES	YES	YES
Year FE	NO	YES	NO	YES	NO	YES
Industry-Year FE	YES	NO	YES	NO	YES	NO
State-Year FE	YES	NO	YES	NO	YES	NO

Notes: *, **, and *** denote $p < 0.10$, $p < 0.05$, and $p < 0.01$, respectively. Standard errors (clustered by firm) are reported in parentheses. The unit of observation is the firm-year. *1 Interlock*, *2 Interlocks* and *3+ Interlocks* are indicators for the number of Section 8–violating interlocks the firm has in year t ; the omitted category is no interlock. All specifications include a set of firm-level controls chosen by double LASSO, the estimated firm-level AKM effects, and the corresponding sets of fixed effects as reported in the table.

Table A.5.2: Effects of Horizontal Interlock Brackets on Innovation Outputs

	(1) Log(Patents)	(2) Log(Patents)	(3) Log(CW Patents)	(4) Log(CW Patents)
1 Interlock	0.088** (0.042)	0.062 (0.038)	0.061 (0.056)	0.056 (0.047)
2 Interlocks	0.140 (0.125)	0.103 (0.099)	0.163 (0.122)	0.086 (0.103)
3+ Interlocks	0.076 (0.265)	0.025 (0.262)	0.297 (0.193)	0.263* (0.148)
N	3,889	4,121	3,933	4,165
R-squared	0.936	0.920	0.898	0.871
Firm FE	YES	YES	YES	YES
Year FE	NO	YES	NO	YES
Industry-Year FE	YES	NO	YES	NO
State-Year FE	YES	NO	YES	NO

Notes: *, **, and *** denote $p < 0.10$, $p < 0.05$, and $p < 0.01$, respectively. Standard errors (clustered by firm) are reported in parentheses. The unit of observation is the firm-year. *1 Interlock*, *2 Interlocks* and *3+ Interlocks* are indicators for the number of Section 8-violating interlocks the firm has in year t ; the omitted category is no interlock. All specifications include a set of firm-level controls chosen by double LASSO, the estimated firm-level AKM effects, and the corresponding sets of fixed effects as reported in the table.

Table A.5.3: Effects of Horizontal Interlock Brackets on Innovation Values

	(1) Log(Total Value)	(2) Log(Total Value)	(3) Log(Avg. Value)	(4) Log(Avg. Value)
1 Interlock	0.143* (0.073)	0.122* (0.065)	0.032 (0.057)	0.035 (0.053)
2 Interlocks	0.235 (0.170)	0.232* (0.136)	0.137 (0.087)	0.089 (0.076)
3+ Interlocks	0.240 (0.334)	-0.051 (0.279)	-0.109 (0.218)	-0.248 (0.197)
N	3,704	3,930	3,237	3,470
R-squared	0.925	0.899	0.853	0.791
Firm FE	YES	YES	YES	YES
Year FE	NO	YES	NO	YES
Industry-Year FE	YES	NO	YES	NO
State-Year FE	YES	NO	YES	NO

Notes: *, **, and *** denote $p < 0.10$, $p < 0.05$, and $p < 0.01$, respectively. Standard errors (clustered by firm) are reported in parentheses. The unit of observation is the firm-year. *1 Interlock*, *2 Interlocks* and *3+ Interlocks* are indicators for the number of Section 8-violating interlocks the firm has in year t ; the omitted category is no interlock. All specifications include a set of firm-level controls chosen by double LASSO, the estimated firm-level AKM effects, and the corresponding sets of fixed effects as reported in the table.

A.6 Markups

A.6.1 Implementation Details for Markup Estimation

I build a firm–year panel and retain observations with non-missing gross revenue, materials expenditure, and the book value of plant, property, and equipment. Inputs are constructed as logs of gross output, materials, and capital and log employment as a labor quantity. I estimate a translog gross-output production function separately within 2-digit industries:

$$\begin{aligned} \ln Q_{it} = & \beta_0 + \beta_M \ln M_{it} + \beta_L \ln L_{it} + \beta_K \ln K_{it} \\ & + \frac{1}{2} \beta_{MM} (\ln M_{it})^2 + \beta_{ML} \ln M_{it} \ln L_{it} + \beta_{MK} \ln M_{it} \ln K_{it} + \dots + \omega_{it} + \varepsilon_{it}. \end{aligned} \quad (\text{A.6.1})$$

To address simultaneity and selection, I use the [Olley and Pakes \(1996\)](#) investment-proxy estimator with the [Akerberg, Caves and Frazer \(2015\)](#) correction to estimate translog coefficients and a firm–year productivity estimate $\hat{\omega}_{it}$ from the first-stage residual. The mapping from elasticities and expenditure shares to markups follows the [De Loecker and Warzynski \(2012\)](#) framework.

Given the estimated translog, the materials elasticity varies by firm and year:

$$\theta_{M,it} = \hat{\beta}_M + 2\hat{\beta}_{MM} \ln M_{it} + \hat{\beta}_{ML} \ln L_{it} + \hat{\beta}_{MK} \ln K_{it}$$

The materials revenue share is

$$\alpha_{M,it} = \frac{\text{materials expenditure}_{it}}{\widehat{P_{it}^Q Q_{it}}}, \quad \widehat{P_{it}^Q Q_{it}} = \frac{\text{gross revenue}_{it}}{\exp(\hat{\omega}_{it})},$$

i.e., I scale revenue by $\exp(\hat{\omega}_{it})$ to attenuate price components that contaminate revenue in production estimation.

Firm–year markups are computed as

$$\mu_{it} = \frac{\theta_{M,it}}{\alpha_{M,it}},$$

as implied by cost minimization with competitive input markets ([De Loecker and Warzyn-](#)

ski, 2012). For each firm–year, I retain $(\mu_{it}, \alpha_{M,it}, \hat{\omega}_{it})$ and merge sectoral estimates back to the full panel.

A.6.2 Markup Results

Table A.6.1 presents the results of running Equations 1 and 2 on the markup numbers estimated using the procedure illustrated in this section. In each specification, the coefficient on firm-level markups is null.

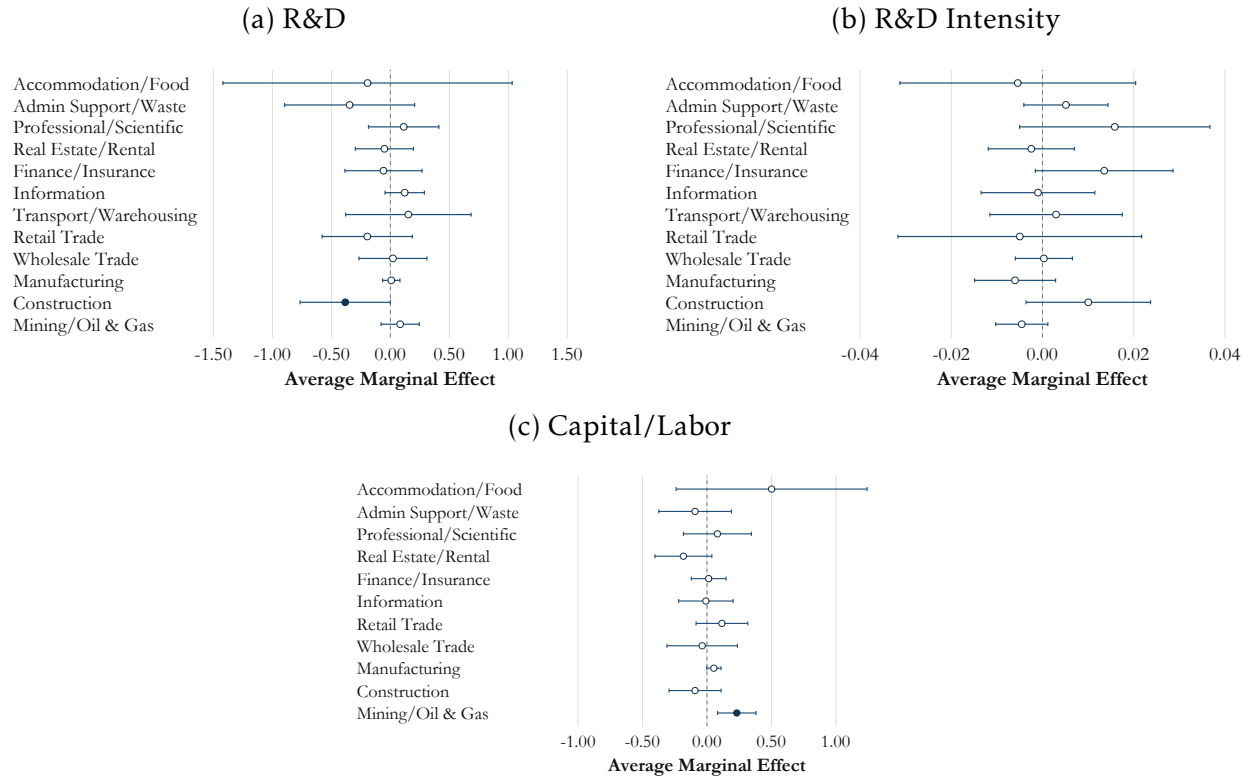
Table A.6.1: Effects of Horizontal Interlocks on Markups

	(1) Log(Markup)	(2) Log(Markup)
Interlock	-0.004 (0.012)	-0.006 (0.010)
N	4,632	4,880
R-squared	0.969	0.961
Firm FE	YES	YES
Year FE	NO	YES
Industry-Year FE	YES	NO
State-Year FE	YES	NO

Notes: *, **, and *** denote $p < 0.10$, $p < 0.05$, and $p < 0.01$, respectively. Standard errors (clustered by firm) are reported in parentheses. The unit of observation is the firm-year. *Interlock* is an indicator equal to one if the firm has at least one Section 8–violating horizontal interlock in year t and zero otherwise. All specifications include a set of firm-level controls chosen by double LASSO, the estimated firm-level AKM effects, and the corresponding sets of fixed effects as reported in the table.

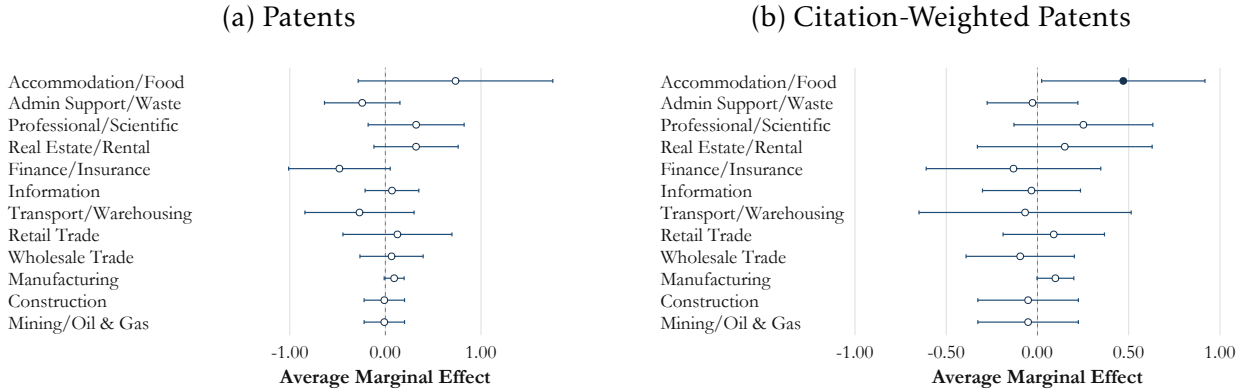
A.7 Results by Industry

Figure A.7.1: Effects of Section 8 Interlocks on Innovation Inputs by Industry



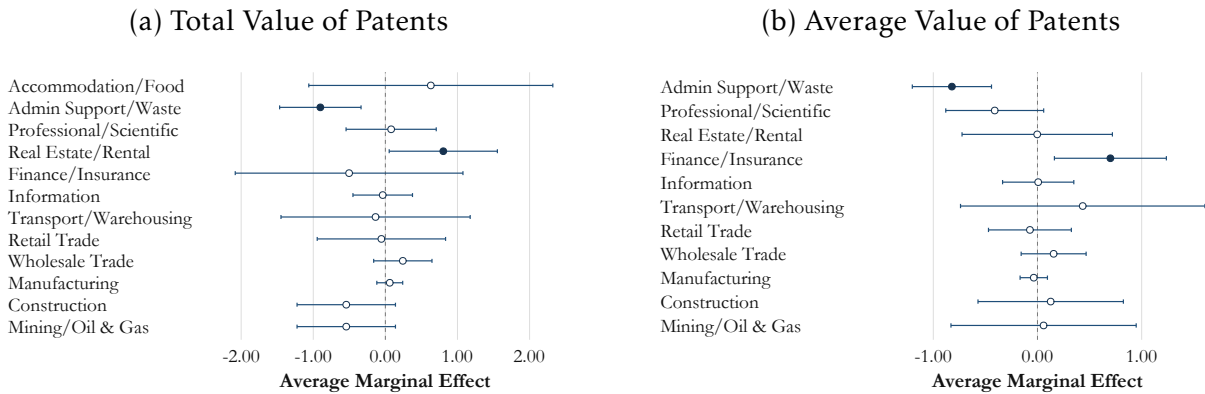
Notes: These graphs report the average marginal effect of Interlock_{jt} estimated from Regression 1 augmented with a full set of interactions between Interlock_{jt} and industry indicators. Each dot denotes the estimated effect for a different industry, defined by NAICS-2 sectors (with codes 31–33, 44–45 and 48–49 collapsed into single sectors). Only industries containing at least one firm whose interlock status changes over the sample period are shown. The vertical navy bars denote the corresponding 95% confidence intervals, based on standard errors clustered at the firm level.

Figure A.7.2: Effects of Section 8 Interlocks on Innovation Outputs by Industry



Notes: These graphs report the average marginal effect of Interlock_{jt} estimated from Regression 1 augmented with a full set of interactions between Interlock_{jt} and industry indicators. Each dot denotes the estimated effect for a different industry, defined by NAICS-2 sectors (with codes 31–33, 44–45 and 48–49 collapsed into single sectors). Only industries containing at least one firm whose interlock status changes over the sample period are shown. The vertical navy bars denote the corresponding 95% confidence intervals, based on standard errors clustered at the firm level.

Figure A.7.3: Effects of Section 8 Interlocks on Innovation Values by Industry



Notes: These graphs report the average marginal effect of Interlock_{jt} estimated from Regression 1 augmented with a full set of interactions between Interlock_{jt} and industry indicators. Each dot denotes the estimated effect for a different industry, defined by NAICS-2 sectors (with codes 31–33, 44–45 and 48–49 collapsed into single sectors). Only industries containing at least one firm whose interlock status changes over the sample period are shown. The vertical navy bars denote the corresponding 95% confidence intervals, based on standard errors clustered at the firm level.